



US012408316B2

(12) **United States Patent**
Fujiwara et al.

(10) **Patent No.:** **US 12,408,316 B2**
(45) **Date of Patent:** ***Sep. 2, 2025**

(54) **MEMORY ARRAY CIRCUIT AND METHOD OF MANUFACTURING SAME**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/756,363**

(22) Filed: **Jun. 27, 2024**

(65) **Prior Publication Data**

US 2024/0357788 A1 Oct. 24, 2024

Related U.S. Application Data

(60) Continuation of application No. 18/304,301, filed on Apr. 20, 2023, now Pat. No. 12,029,023, which is a (Continued)

(51) **Int. Cl.**
H10B 10/00 (2023.01)
G06F 30/392 (2020.01)
(Continued)

(52) **U.S. Cl.**
CPC **H10B 10/12** (2023.02); **G06F 30/392** (2020.01); **H01L 23/5226** (2013.01); **H01L 23/5286** (2013.01); **H10D 89/10** (2025.01)

(58) **Field of Classification Search**

CPC H10B 10/12; H10B 43/00; H10B 43/30; G06F 30/392; H01L 23/5226;
(Continued)

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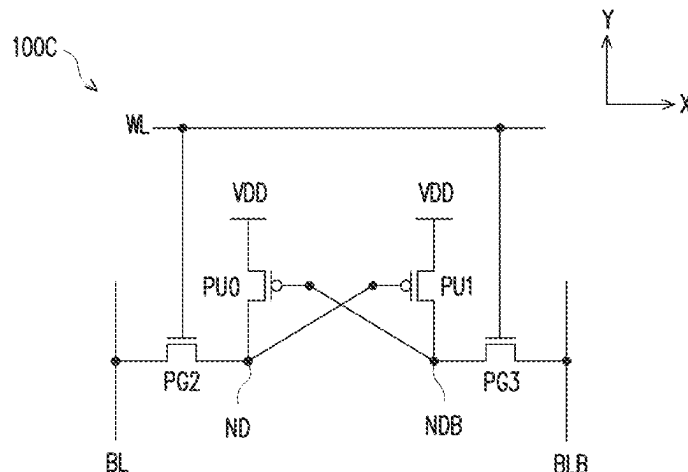
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(57) **ABSTRACT**

A memory circuit includes a first pull down transistor, a first pass gate transistor coupled to the first pull down transistor, a second pull down transistor, a second pass gate transistor and a first metal contact. The second pull down transistor has a first active region located on a first level. The second pass gate transistor has a second active region located on the first level, and being coupled to the second pull down transistor. The first metal contact extends from the first active region to the second active region, being located on a second level, and electrically coupling a drain of the second pull down transistor to a drain of the second pass gate transistor. The first pass gate transistor, the second pass gate transistor, the first pull down transistor and the second pull down transistor are part of a four transistor (4T) memory cell.

20 Claims, 17 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/325,641, filed on May 20, 2021, now Pat. No. 11,637,108, which is a division of application No. 16/457,553, filed on Jun. 28, 2019, now Pat. No. 11,018,142.

(60) Provisional application No. 62/698,665, filed on Jul. 16, 2018.

(51) **Int. Cl.**

H01L 23/522 (2006.01)

H01L 23/528 (2006.01)

H10D 89/10 (2025.01)

(58) **Field of Classification Search**

CPC ... H01L 23/5286; H10D 89/10; G11C 11/413;
G11C 11/412; F27D 17/25

USPC 257/401

See application file for complete search history.

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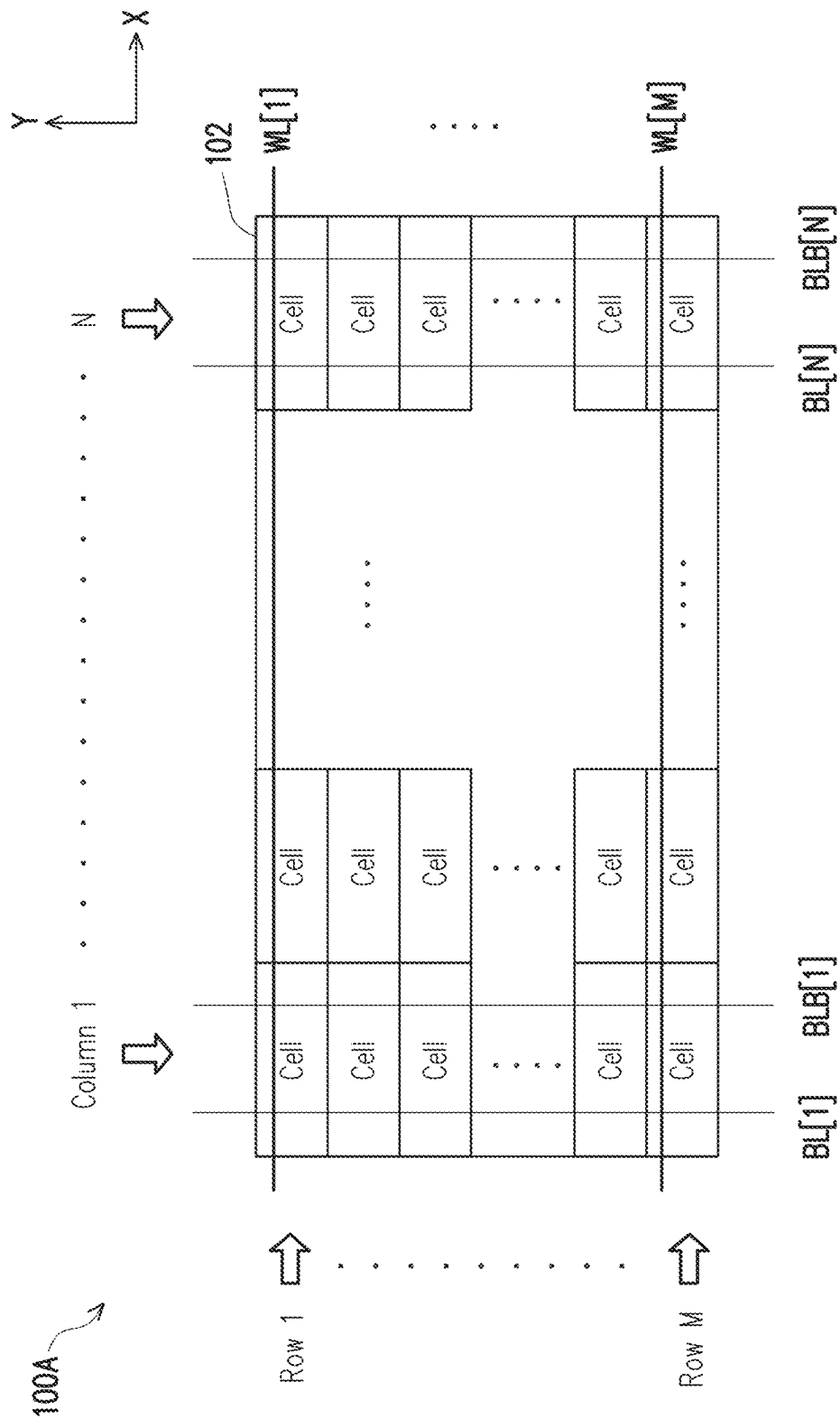


FIG. 1A

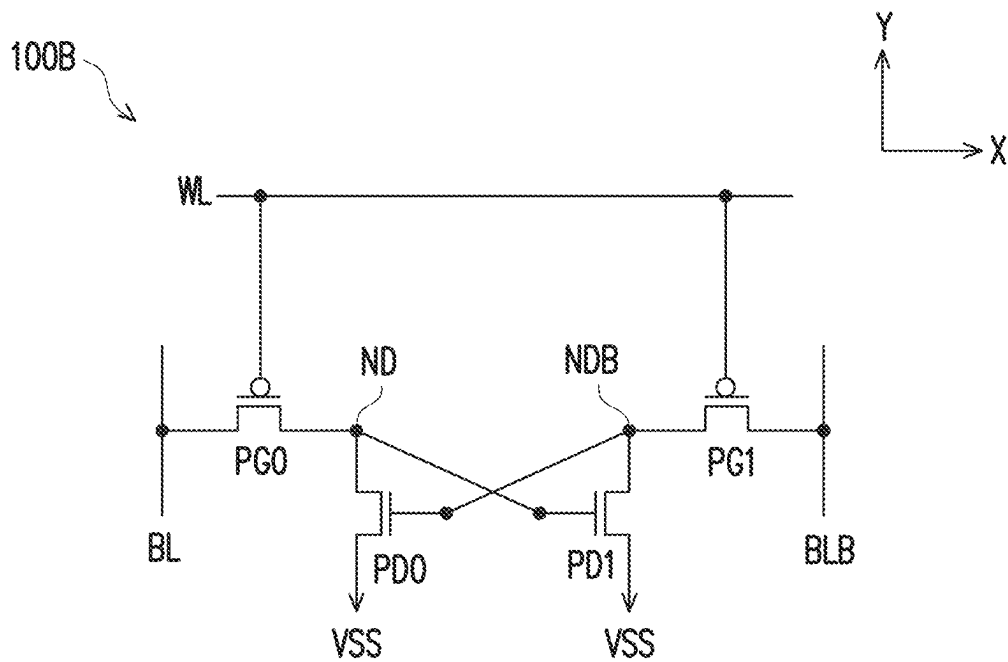


FIG. 1B

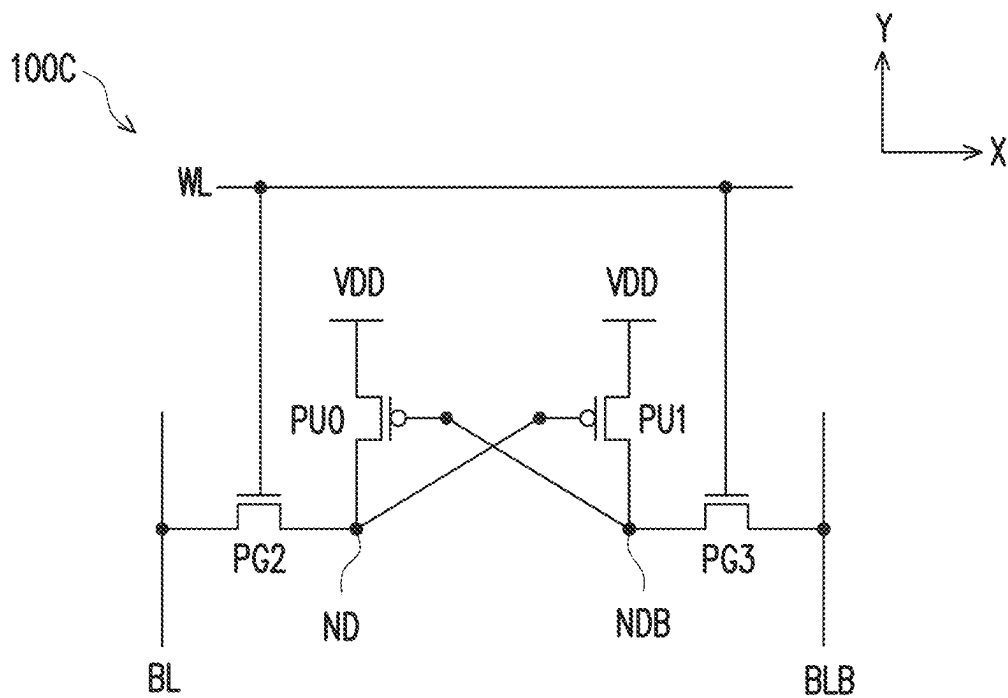
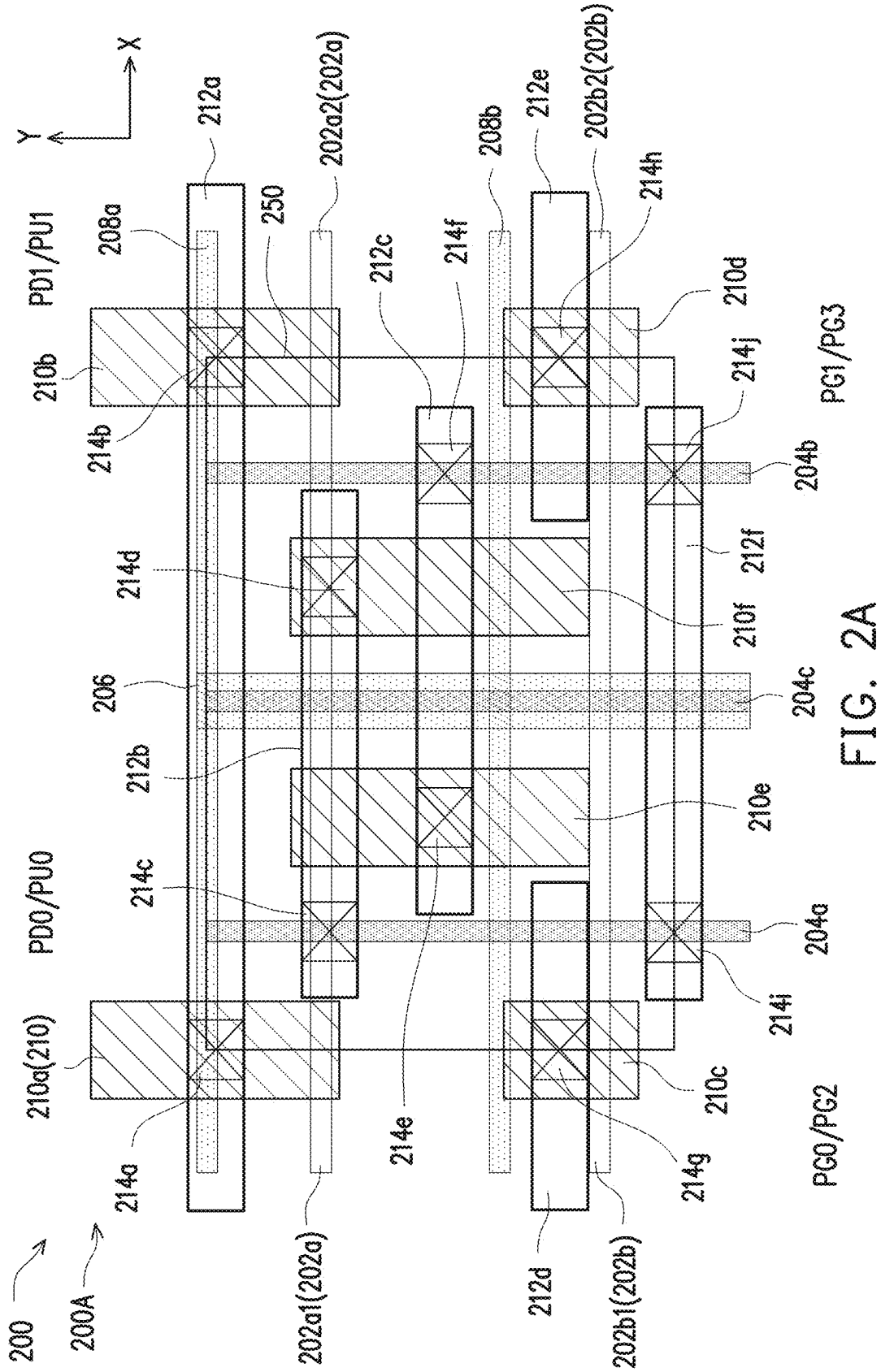


FIG. 1C



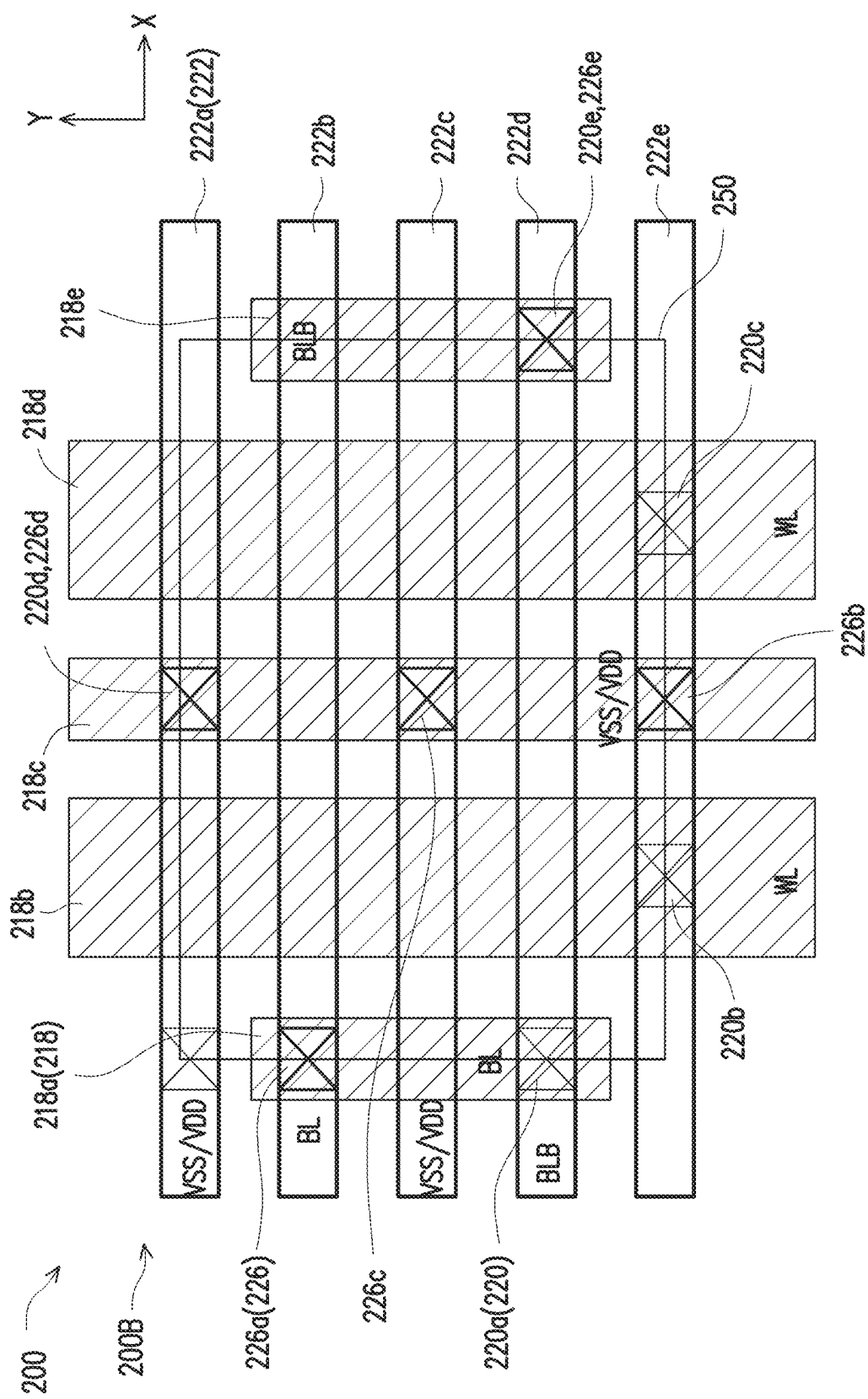


FIG. 2B

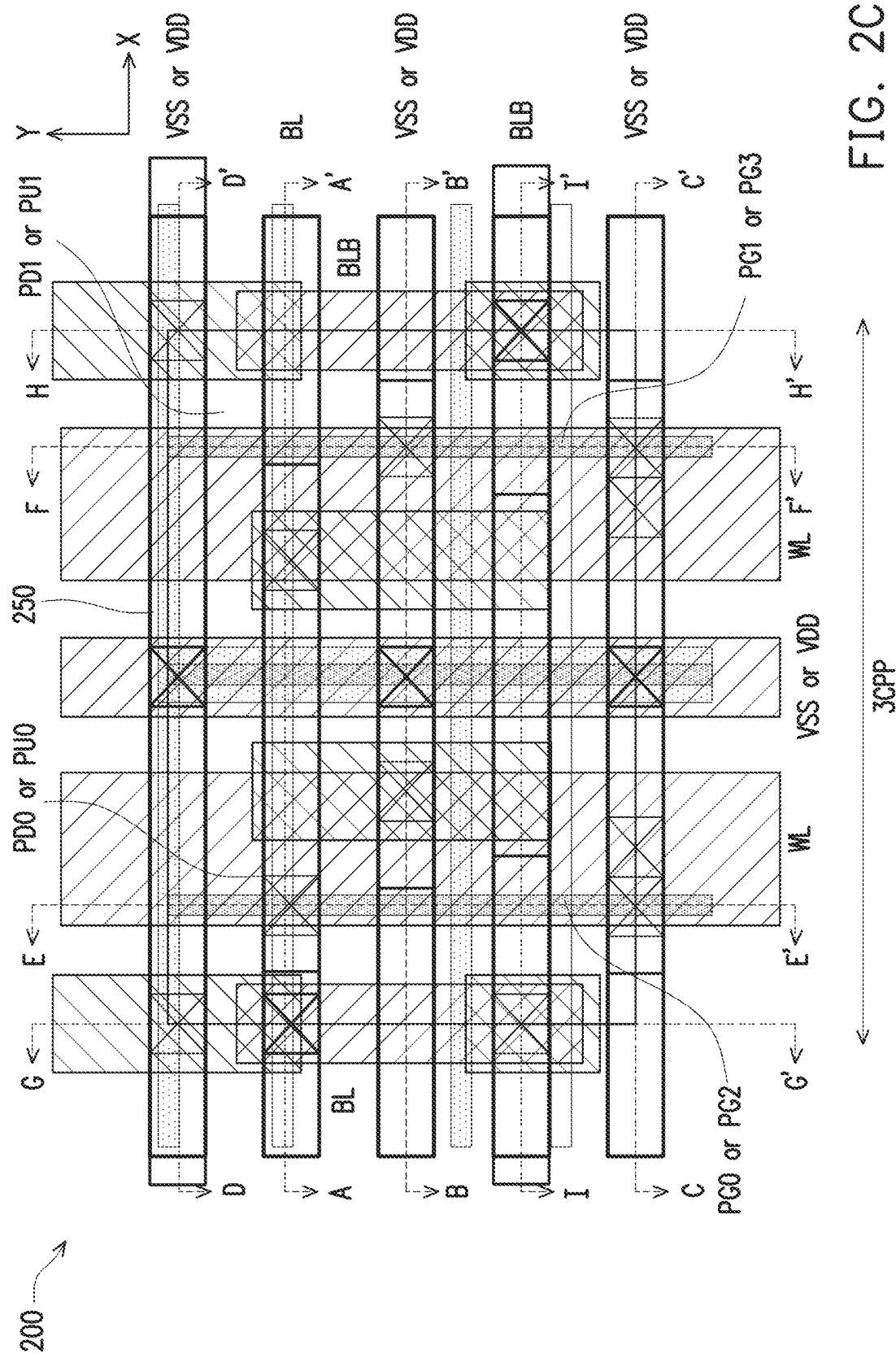


FIG. 2C

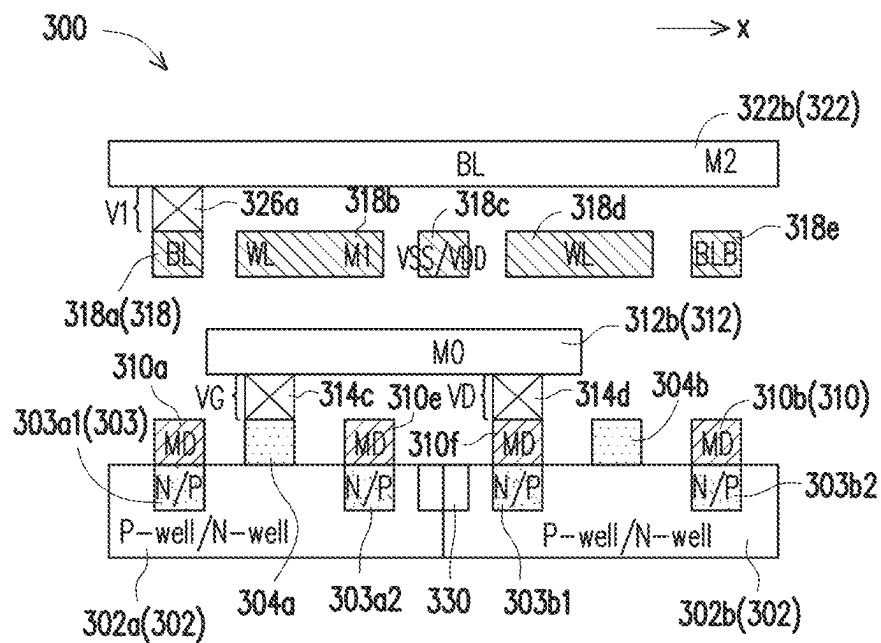


FIG. 3A

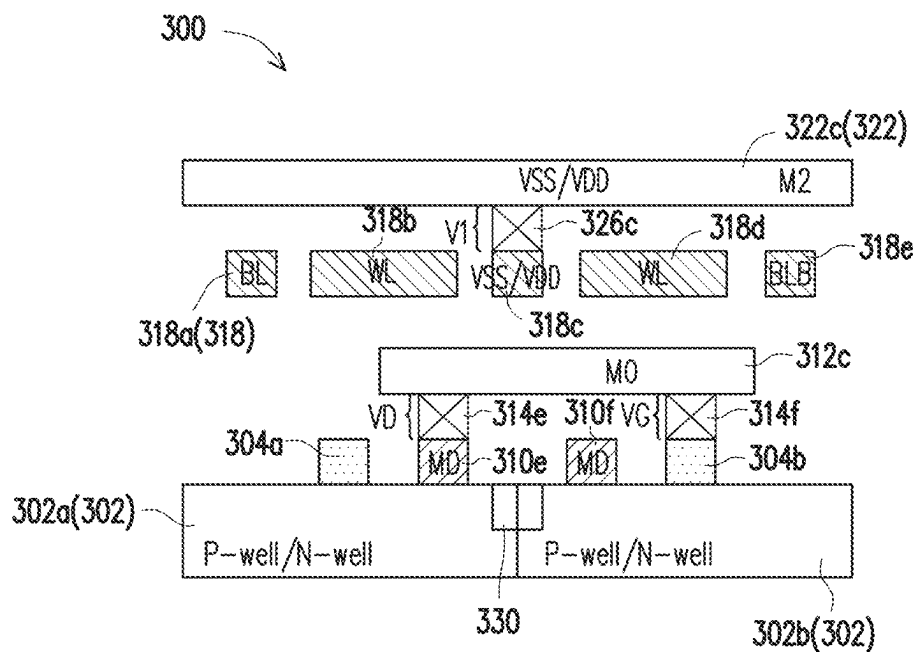


FIG. 3B

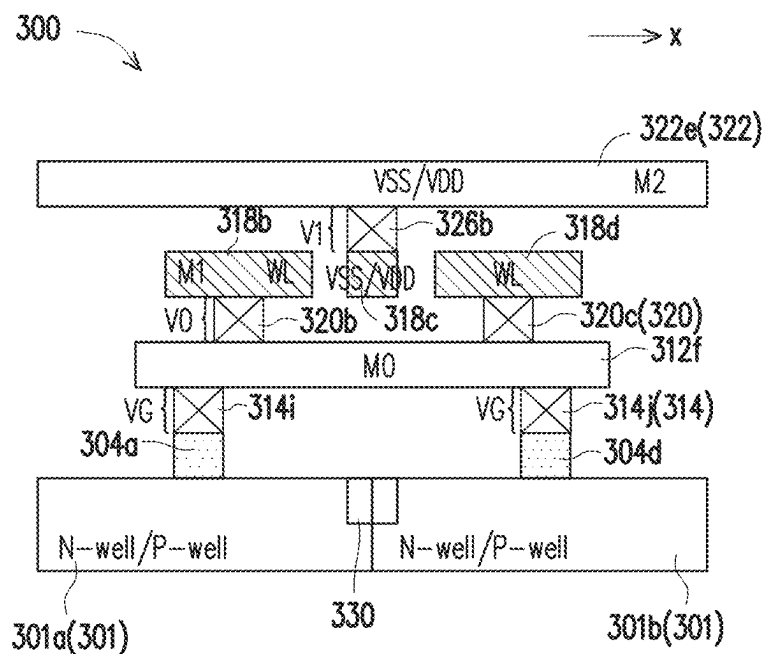


FIG. 3C

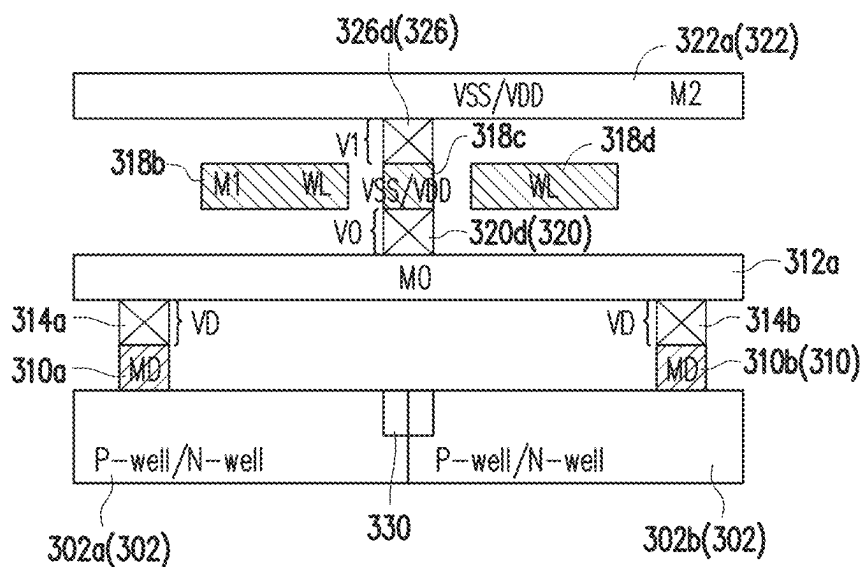


FIG. 3D

FIG. 3F

FIG. 3H

FIG. 31

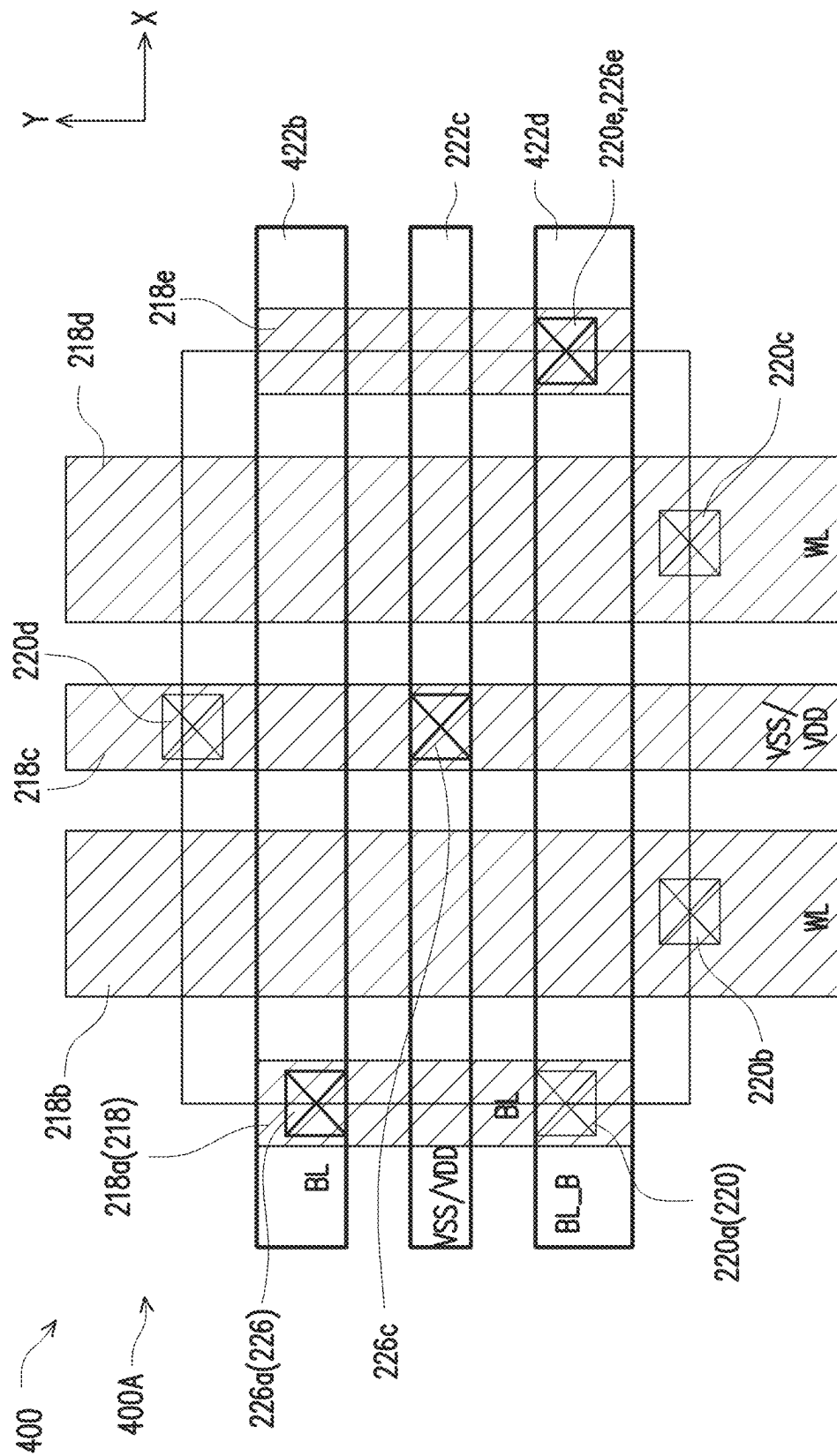


FIG. 4A

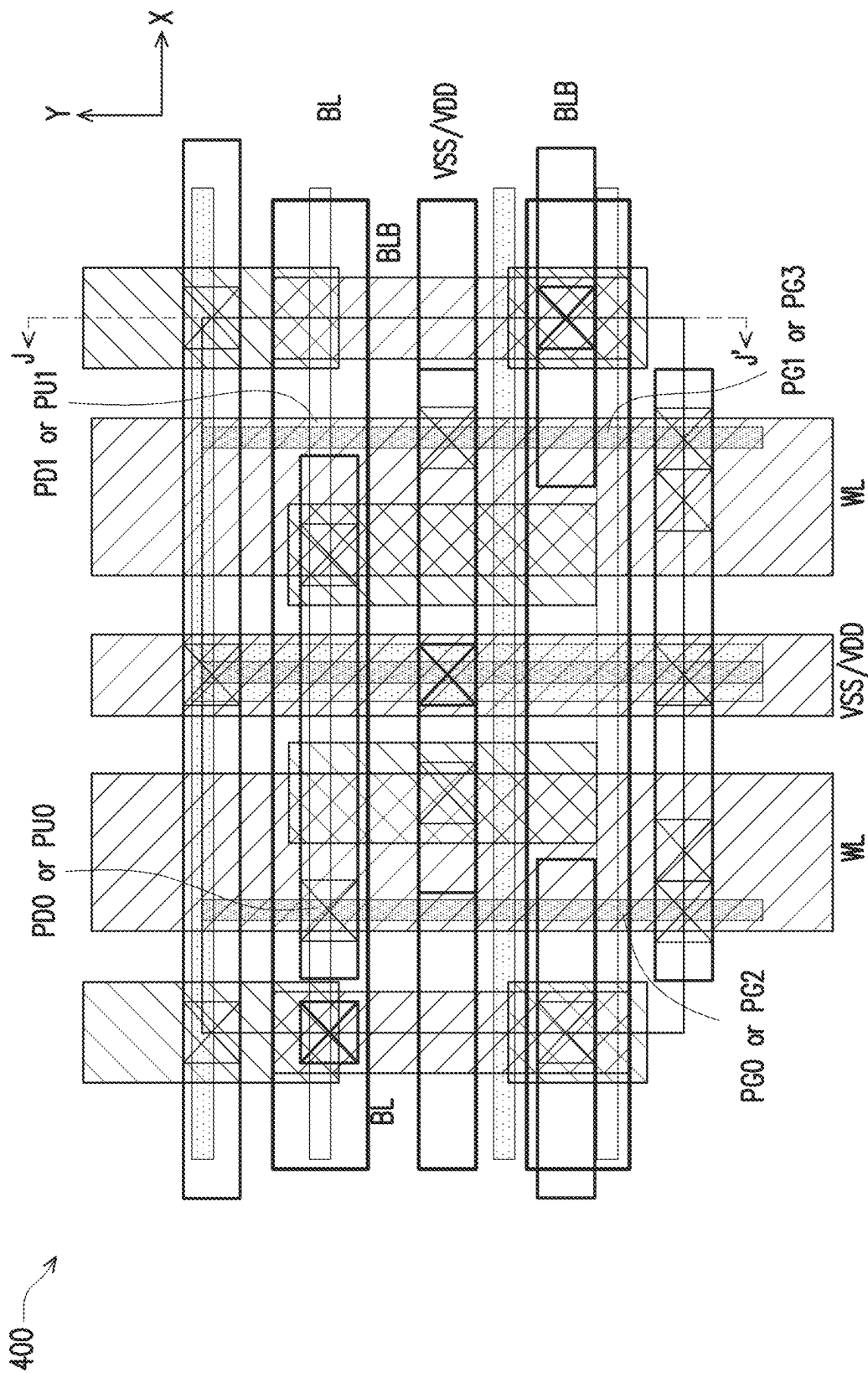


FIG. 4B

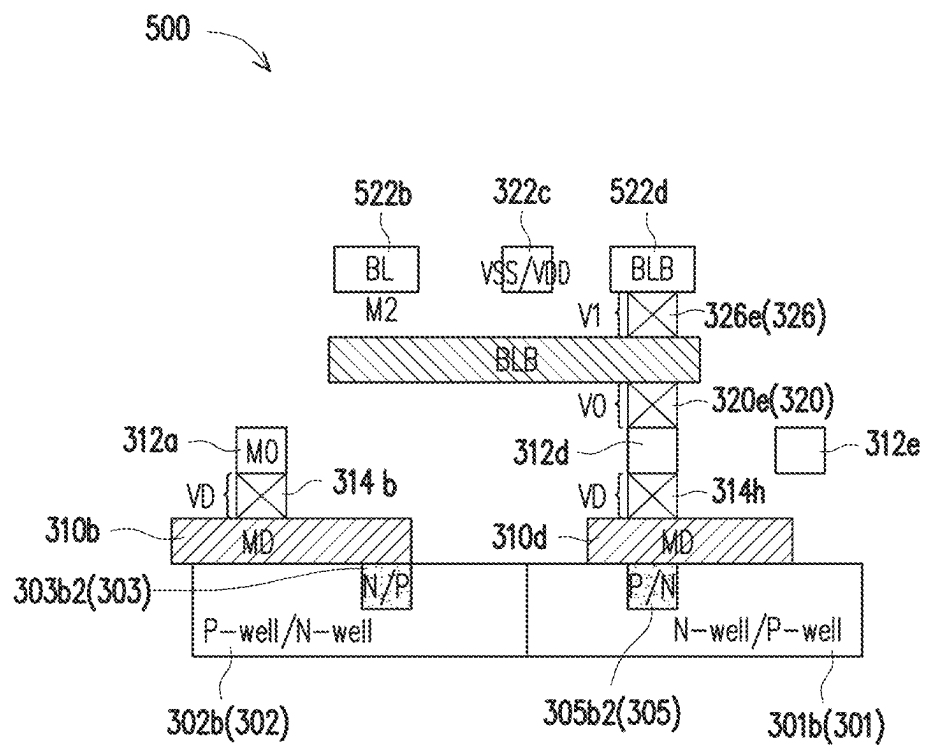


FIG. 5

600A

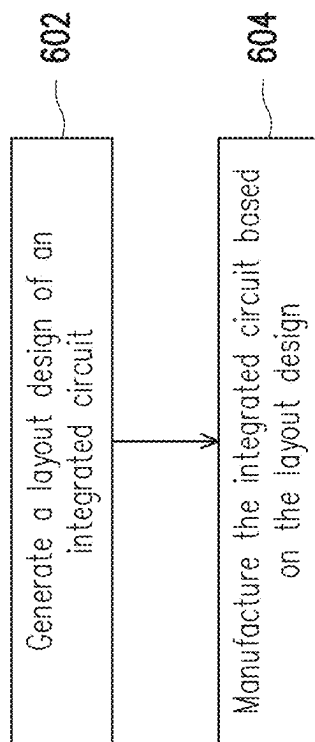


FIG. 6A

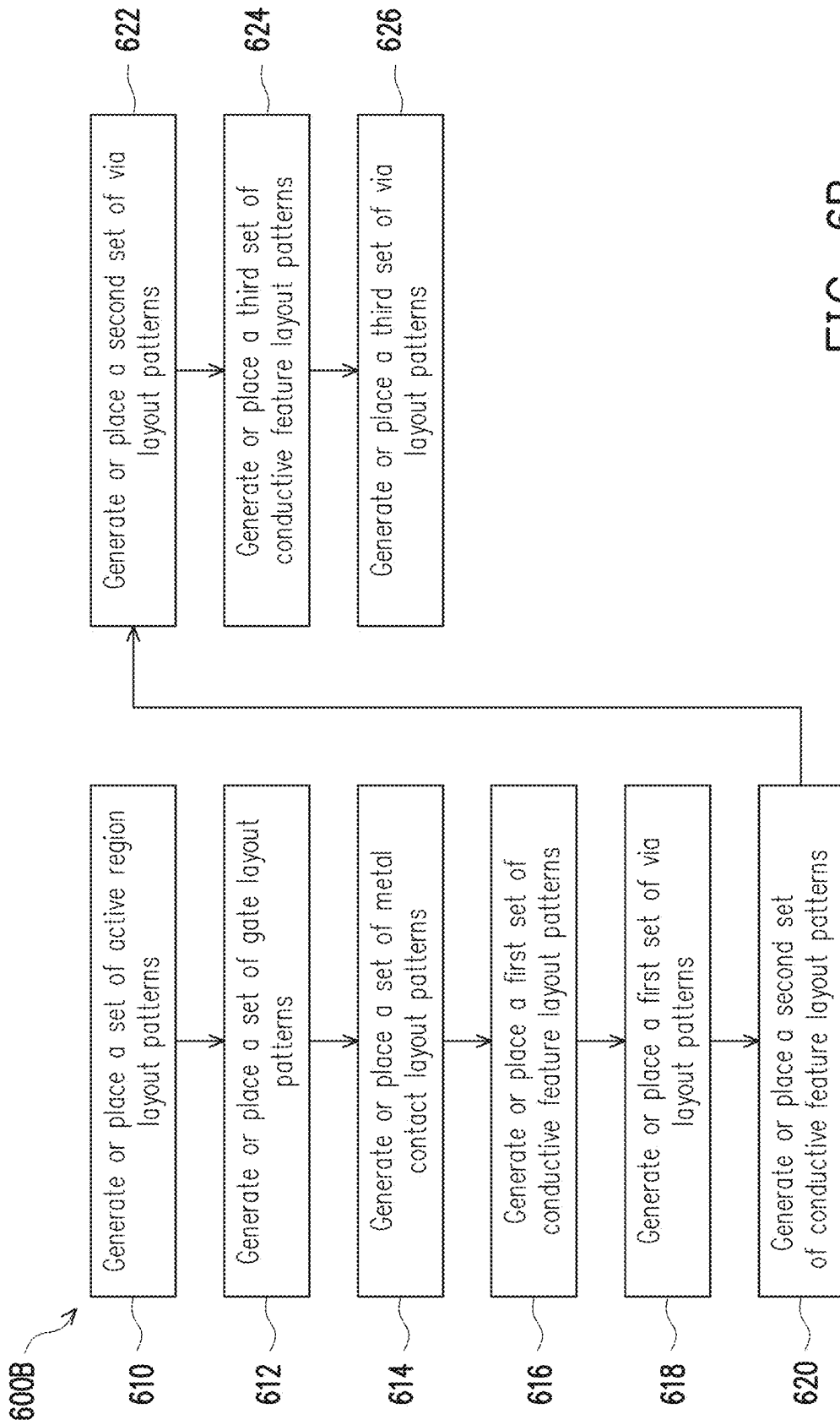


FIG. 6B

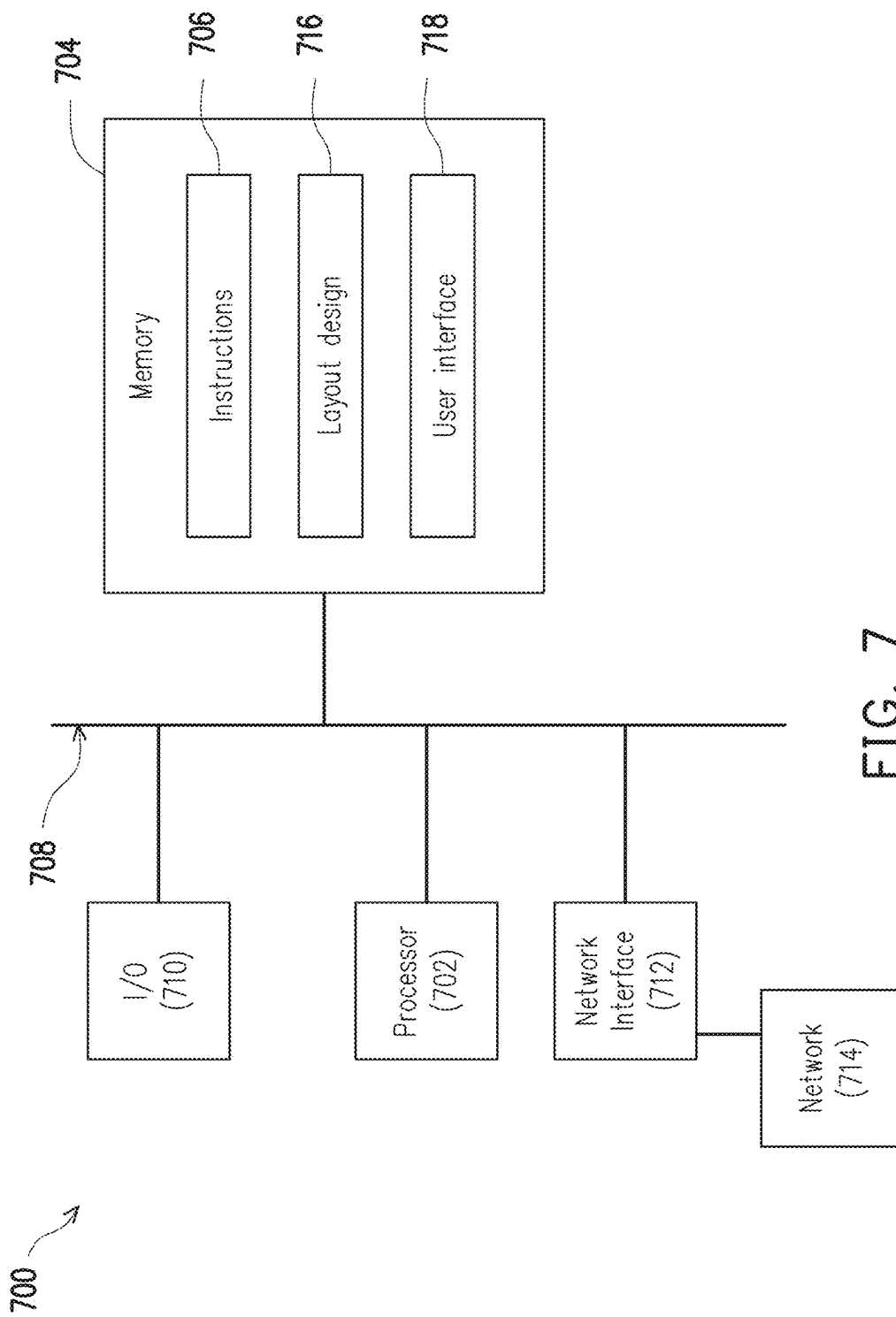


FIG. 7

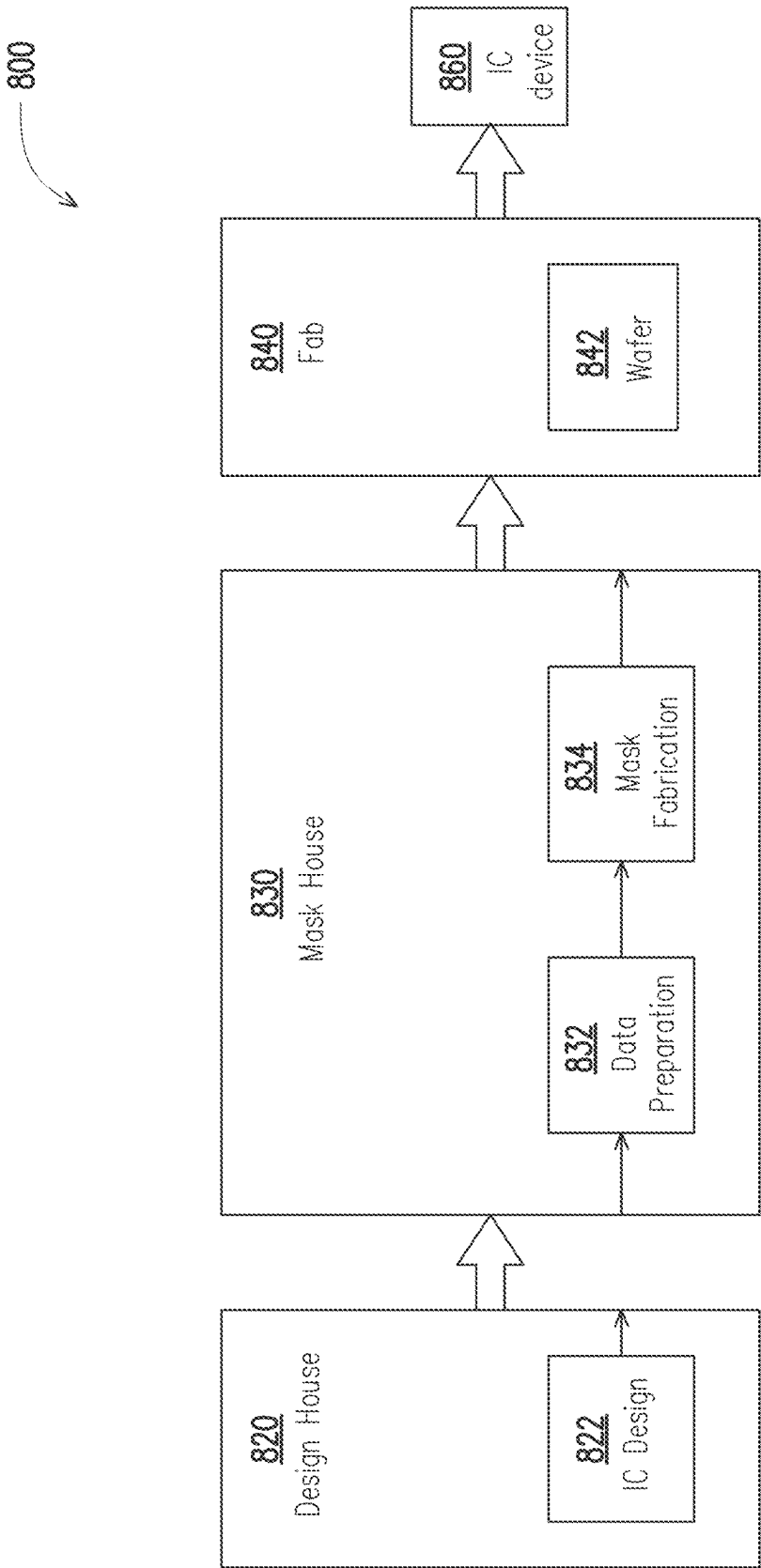


FIG. 8

MEMORY ARRAY CIRCUIT AND METHOD OF MANUFACTURING SAME

PRIORITY CLAIM AND CROSS-REFERENCE

The present application is a continuation of U.S. application Ser. No. 18/304,301 filed Apr. 20, 2023, now U.S. Pat. No. 12,029,023, issued Jul. 2, 2024, which is a continuation of U.S. application Ser. No. 17/325,641 filed May 20, 2021, now U.S. Pat. No. 11,637,108, issued Apr. 25, 2023, which is a divisional of U.S. application Ser. No. 16/457,553 filed Jun. 28, 2019, now U.S. Pat. No. 11,018,142, issued May 25, 2021, which claims the benefit of U.S. Provisional Application No. 62/698,665, filed Jul. 16, 2018, each of which are incorporated herein by reference in their entireties.

BACKGROUND

The semiconductor integrated circuit (IC) industry has produced a wide variety of digital devices to address issues in a number of different areas. Some of these digital devices, such as memory macros, are configured for the storage of data. As ICs have become smaller and more complex, the resistance of conductive lines within these digital devices are also changed affecting the operating voltages of these digital devices and overall IC performance.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1A is a circuit diagram of a memory macro, in accordance with some embodiments.

FIG. 1B is a circuit diagram of a memory cell, in accordance with some embodiments.

FIG. 1C is a circuit diagram of a memory cell, in accordance with some embodiments.

FIGS. 2A, 2B and 2C are diagrams of a layout design, in accordance with some embodiments.

FIGS. 3A, 3B, 3C, 3D, 3E, 3F, 3G, 3H and 3I are diagrams of at least one integrated circuit, in accordance with some embodiments.

FIGS. 4A and 4B are diagrams of a layout design, in accordance with some embodiments.

FIG. 5 is a diagram of an integrated circuit, in accordance with some embodiments.

FIG. 6A is a flowchart of a method of manufacturing an integrated circuit, in accordance with some embodiments.

FIG. 6B is a flowchart of a method of generating a layout design of a memory array circuit, in accordance with some embodiments.

FIG. 7 is a block diagram of a system for designing an integrated circuit layout design, in accordance with some embodiments.

FIG. 8 is a block diagram of an integrated circuit (IC) manufacturing system and an IC manufacturing flow associated therewith, in accordance with some embodiments.

DETAILED DESCRIPTION

The following disclosure provides different embodiments, or examples, for implementing features of the provided

subject matter. Specific examples of components, materials, values, steps, arrangements, or the like, are described below to simplify the present disclosure. These are, of course, merely examples and are not limiting. Other components, materials, values, steps, arrangements, or the like, are contemplated. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

In accordance with some embodiments, a memory cell includes a first and second pull up transistor, a first and second pass gate transistor and a first metal contact. In some embodiments, the first pass gate transistor, the second pass gate transistor, the first pull up transistor and the second pull up transistor are part of a four transistor (4T) memory cell. In some embodiments, the second pass gate transistor is coupled to the second pull up transistor.

In some embodiments, the first pull up transistor has a first active region extending in a first direction, and is located on a first level. In some embodiments, the first pass gate transistor has a second active region extending in the first direction. In some embodiments, the second active region is located on the first level, and is separated from the first active region in a second direction different from the first direction. In some embodiments, the second active region is adjacent to the first active region.

In some embodiments, the first metal contact extends in the second direction, and extends from the first active region to the second active region. In some embodiments, the first metal contact is located on a second level different from the first level. In some embodiments, the first metal contact electrically couples a drain of the first pull up transistor to a drain of the first pass gate transistor.

Memory Macro

FIG. 1A is a circuit diagram of a memory macro 100A, in accordance with some embodiments. In the embodiment of FIG. 1, memory macro 100A is a static random access memory (SRAM) macro. SRAM is used for illustration, and other types of memories are within the scope of various embodiments.

Memory macro 100A comprises an array of cells 102 having M rows and N columns, where N is a positive integer corresponding to the number of columns in array of cells 102 and M is a positive integer corresponding to the number of rows in array of cells 102. The rows of cells in array of cells 102 are arranged in a first direction X. The columns of cells in array of cells 102 are arranged in a second direction Y. The second direction Y is different from the first direction X.

In some embodiments, the second direction is perpendicular to the first direction. In some embodiments, array of cells **102** includes one or more single port (SP) SRAM cells. In some embodiments, array of cells **102** includes one or more dual port (DP) SRAM cells. Different types of memory cells in array of cells **102** are within the contemplated scope of the present disclosure.

Memory macro **100A** further includes N bit lines BL[1], . . . BL[N] (collectively referred to as “bit line BL”) and N bit line bars BLB[1], . . . BLB[N] (collectively referred to as “bit line bar BLB”). Each column 1, . . . , N in array of cells **102** is overlapped by a corresponding bit line BL[1], . . . , BL[N] and a corresponding bit line bar BLB[1], . . . , BLB[N]. Each bit line BL or bit line bar BLB extends in the second direction Y and over a column of cells (e.g., column 1, . . . , N).

Memory macro **100A** further includes M word lines WL[1], . . . WL[M] (collectively referred to as “word line WL”). Each row 1, . . . , M in array of cells **102** is overlapped by a corresponding word line WL[1], . . . , WL[M]. Each word line WL extends in the first direction X and over a corresponding row of cells (e.g., row 1, . . . , M).

Different configurations of memory macro **100A** are within the contemplated scope of the present disclosure.

A Memory Cell

FIG. 1B is a circuit diagram of a memory cell **100B** useable in FIG. 1A, in accordance with some embodiments.

Memory cell **100B** is usable as one or more memory cells in memory macro **100A** of FIG. 1A.

Memory cell **100B** is a four transistor (4T) single port (SP) SRAM memory cell used for illustration. In some embodiments, memory cell **100B** includes a number of transistors other than four. Other types of memory are within the scope of various embodiments.

Memory cell **100B** comprises two P-type metal oxide semiconductor (PMOS) transistors PG0 and PG1, and two N-type metal oxide semiconductor (NMOS) transistors PD0 and PD1. PMOS transistors includes and PG1 are configured as pass-gate transistors, and NMOS transistors PD0 and PD1 are configured as pull-down transistors.

A drain terminal of NMOS transistor PD0, a gate terminal of NMOS transistor PD1, and a source terminal of PMOS transistor PG0 are coupled together at a node configured as a storage node ND. A drain terminal of NMOS transistor PD1, a gate terminal of NMOS transistor PD0, and a source terminal of PMOS transistor PG1 are coupled together at a node configured as a storage node NDB.

A source terminal of each of NMOS transistors PD0 and PD1 is configured as a supply reference voltage node (not labelled) having a supply reference voltage VSS. The source terminal of each of NMOS transistors PD0 and PD1 is also coupled to supply reference voltage VSS.

A word line WL is coupled with a gate terminal of each of PMOS transistors PG0 and PG1. Word line WL is also called a write control line because PMOS transistors PG0 and PG1 are configured to be controlled by a signal on word line WL in order to transfer data between bit lines BL, BLB and corresponding nodes ND, NDB.

A drain terminal of PMOS transistor PG0 is coupled to a bit line BL. A drain terminal of PMOS transistor PG1 is coupled to a bit line BLB. Bit lines BL and BLB are configured as both data input and output for memory cell **100B**. In some embodiments, in a write operation, applying a logical value to a first bit line BL and the opposite logical value to the other bit line BLB enables writing the logical values on the bit lines to memory cell **100B**. Each of bit lines BL and BLB is called a data line because the data carried on

bit lines BL and BLB are written to and read from corresponding nodes ND and NDB.

Different configurations of memory cell **100B** are within the contemplated scope of the present disclosure. For example, source or drain terminals of PMOS transistor PG0, PMOS transistor PG1, NMOS transistor PD0, or NMOS transistor PD1 can be swapped for corresponding drain or source terminals of PMOS transistor PG0, PMOS transistor PG1, NMOS transistor PD0, or NMOS transistor PD1, and vice versa.

Another Memory Cell

FIG. 1C is a circuit diagram of a memory cell **100C** useable in FIG. 1A, in accordance with some embodiments.

Memory cell **100C** is usable as one or more memory cells in memory macro **100A** of FIG. 1A.

Memory cell **100C** is a 4T SP SRAM memory cell used for illustration. In some embodiments, memory cell **100C** includes a number of transistors other than four. Other types of memory are within the scope of various embodiments.

Memory cell **100C** comprises two PMOS transistors PU0 and PU1, and two NMOS transistors PG2 and PG3. NMOS transistors PG2 and PG3 are configured as pass-gate transistors, and PMOS transistors PU0 and PU1 are configured as pull-up transistors.

A drain terminal of PMOS transistor PU0, a gate terminal of PMOS transistor PU1, and a source terminal of NMOS transistor PG2 are coupled together at a node configured as storage node ND. A drain terminal of PMOS transistor PU1, a gate terminal of PMOS transistor PU0, and a source terminal of NMOS transistor PG3 are coupled together at a node configured as storage node NDB.

A source terminal of each of PMOS transistors PU0 and PU1 is configured as a supply voltage node (not labelled) having a supply voltage VDD. The source terminal of each of PMOS transistors PU0 and PU1 is also coupled to supply voltage VDD.

A word line WL is coupled with a gate terminal of each of NMOS transistors PG2 and PG3. NMOS transistors PG2 and PG3 are configured to be controlled by a signal on word line WL in order to transfer data between bit lines BL, BLB and corresponding nodes ND, NDB.

A drain terminal of NMOS transistor PG2 is coupled to a bit line BL. A drain terminal of NMOS transistor PG3 is coupled to a bit line BLB. Bit lines BL and BLB are configured as both data input and output for memory cell **100C**. In some embodiments, in a write operation, applying a logical value to a bit line BL and the opposite logical value to the other bit line BLB enables writing the logical values on the bit lines to memory cell **100C**. Each of bit lines BL and BLB is called a data line because the data carried on bit lines BL and BLB are written to and read from corresponding nodes ND and NDB.

Different configurations of memory cell **100C** are within the contemplated scope of the present disclosure. For example, source or drain terminals of NMOS transistor PG2, NMOS transistor PG3, PMOS transistor PU0 or PMOS transistor PU1 can be swapped for corresponding drain or source terminals of NMOS transistor PG2, NMOS transistor PG3, PMOS transistor PU0 or PMOS transistor PU1, and vice versa.

Layout Design

FIGS. 2A-2C are diagrams of a layout design **200**, in accordance with some embodiments. Layout design **200** is a layout diagram of memory cell **100B** of FIG. 1B or memory cell **100C** of FIG. 1C. Layout design **200** is usable to manufacture memory cell **100B** or **100C**.

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Layout design **200** includes a portion **200A** (FIG. 2A) and a portion **200B** (FIG. 2B). For case of illustration, layout design **200** of FIG. 2A does not include portion **200B**. Similarly, for case of illustration, layout design **200** of FIG. 2B does not include portion **200A**.

Layout design **200**, as shown in FIG. 2C, includes portion **200A** of FIG. 2A and portion **200B** of FIG. 2B. In other words, layout design of FIG. 2C is the combination of layout portion **200A** of FIG. 2A and layout portion **200B** of FIG. 2B, when the cell boundaries of portion **200A** and portion **200B** are aligned. For case of illustration, some of the labeled elements of FIGS. 2A-2B are not labelled in FIG. 2C. In some embodiments, layout design **200** of FIGS. 2A-2C includes additional elements not shown in FIGS. 2A-2C.

Layout design **200** of FIG. 2A includes portion **200A**. Portion **200A** includes features of the active (OD) level, poly (Poly) level, metal on diffusion (MD) level, via over diffusion (VD) level and via over gate (VG) level of layout design **200**.

Layout design **200** of FIG. 2B includes portion **200B**. Portion **200B** includes features of the metal 1 (M1) level, metal two (M2) level, via zero (V0) level and via one (V1) level of layout design **200**.

Layout design **200** includes active region layout patterns **202a** and **202b** (collectively referred to as “set of active region layout patterns **202**”).

Active region layout pattern **202a** is useable to manufacture active region **303a1**, **303a2**, **303b1** and **303b2** of integrated circuit **300** (FIGS. 3A-3I). Active region layout pattern **202b** is useable to manufacture active regions **305a1**, **305a2**, **305b1** and **305b2** of integrated circuit **300** (FIGS. 3A-3I).

Active region layout pattern **202a** includes an active region layout pattern **202a1** and an active region layout pattern **202a2**. In some embodiments, active region layout pattern **202a1** is useable to manufacture active region **303a1** and **303a2** of integrated circuit **300** (FIGS. 3A-3I). In some embodiments, active region layout pattern **202a2** is useable to manufacture active region **303b1** and **303b2** of integrated circuit **300** (FIGS. 3A-3I).

Active region layout pattern **202b** includes an active region layout pattern **202b1** and an active region layout pattern **202b2**. In some embodiments, active region layout pattern **202b1** is useable to manufacture active region **305a1** and **305a2** of integrated circuit **300** (FIGS. 3A-3I). In some embodiments, active region layout pattern **202a2** is useable to manufacture active region **305b1** and **305b2** of integrated circuit **300** (FIGS. 3A-3I).

Each of the layout patterns of the set of active region layout patterns **202a** extends in a first direction X and is located on a first layout level. In some embodiments, the first layout level corresponds to the active region of layout design **200** or **400** (FIG. 4). Layout patterns **202a** and **202b** of the set of active region layout patterns **202** are separated from each other in a second direction Y. In some embodiments, the second direction Y is different from the first direction X. In some embodiments, the set of active region layout patterns **202** is referred to as an oxide definition (OD) layout pattern which defines source or drain diffusion layout patterns of layout design **200** or **400**. In some embodiments, the set of active region layout patterns **202** extends continuously through the cell boundaries of layout design **200** to other neighboring cells.

Layout design **200** further includes gate layout patterns **204a**, **204b** and **204c** (collectively referred to as “set of gate layout patterns **204**”). In some embodiments, gate layout

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patterns **204a** and **204b** are useable to manufacture corresponding gate structures **304a** and **304b** of integrated circuit **300** (FIGS. 3A-3I) and **500** (FIG. 5). Gate layout pattern **204c** is between **204a** and **204b**. In some embodiments, gate layout pattern **204c** is useable to manufacture a dummy gate structure (not shown) of integrated circuit **300** (FIGS. 3A-3I) and **500** (FIG. 5). In some embodiments, a dummy gate structure is a non-functional gate structure. At least the set of gate layout patterns **204** of layout design **200** or **400** (FIGS. 4A-4B) or the set of gates **304** of integrated circuit **300** (FIGS. 3A-3I) and **500** (FIG. 5) have a contact poly pitch (CPP) of 3. Stated differently, in some embodiments, a width of the cell boundary of layout design **200** in the first direction X is equal to the CPP (e.g., 3). In some embodiments, by having a CPP of 3, layout design **200** or **400** (FIGS. 4A-4B) or integrated circuit **300** (FIGS. 3A-3I) and **500** (FIG. 5) have a higher density than other approaches.

In some embodiments, gate layout pattern **204a** is useable to manufacture gate regions of PMOS transistor PG0 and NMOS transistor PD0 or gate regions of PMOS transistor PU0 and NMOS transistor PG2. In some embodiments, gate layout pattern **204b** is useable to manufacture gate regions of PMOS transistor PG1 and NMOS transistor PD1 or gate regions of PMOS transistor PU1 and NMOS transistor PG3.

In some embodiments, each gate layout pattern of the set of gate layout patterns **204** extends in the second direction Y and overlaps the set of active region layout patterns **202**. In some embodiments, each gate layout pattern of the set of gate layout patterns **204** is separated from an adjacent gate layout pattern of the set of gate layout patterns **204** in the first direction X. In some embodiments, an adjacent element is next to or directly next to another element. For example, in some embodiments, gate layout pattern **204a** is adjacent to gate layout pattern **204c**. In some embodiments, active region layout pattern **202a** is adjacent to active region layout pattern **202b**.

The set of gate layout patterns **204** is positioned on a second layout level different from the first layout level. In some embodiments, the second layout level corresponds to the POLY level of layout design **200** or **400** (FIG. 4). The set of active region layout patterns **202** is below the set of gate layout patterns **204**. Other quantities or configurations of the set of gate layout patterns **204** are within the scope of the present disclosure.

Layout design **200** further includes a continuous polysilicon on oxide diffusion (OD) edge (CPODE) layout pattern **206**. CPODE layout pattern **206** extends in the second direction Y, and covers gate layout pattern **204c**. In some embodiments, CPODE layout pattern **206** is useable to indicate that a dummy gate structure (gate structure **304c** manufactured by gate layout pattern **204c**) of integrated circuit **300** (FIGS. 3A-3I) and **500** (FIG. 5) is removed, and a trench is formed and filled with an insulating portion **330** in the first well **301** (FIGS. 3A-3I) and the second well **302** (FIGS. 3A-3I). In some embodiments, a dummy gate structure is a non-functional gate structure. In some embodiments, gate layout pattern **204c** is CPODE layout pattern **206**. In some embodiments, CPODE layout pattern **206** is used to indicate that gate layout pattern **204c** is a dummy gate layout pattern. Other configurations or quantities of patterns in the CPODE layout pattern **206** are within the scope of the present disclosure.

Layout design **200** further includes poly cut feature layout patterns **208a** and **208b** (collectively referred to as “set of cut feature layout patterns **208**”). Set of poly cut feature layout patterns **208** extends in the first direction X. Poly cut feature layout pattern **208b** overlaps set of gate layout patterns **204**

in a middle portion of layout design **200**. Poly cut feature layout pattern **208a** overlaps set of gate layout patterns **204** along cell boundary **250** of layout design **200**. In some embodiments, each cut feature layout pattern (**208a** or **208b**) of the set of poly cut feature layout patterns **208** is separated from another cut feature layout pattern (**208b** or **208a**) of the set of poly cut feature layout patterns **208** in the second direction Y. In some embodiments, the set of poly cut feature layout patterns **208** extends continuously through the cell boundaries of layout design **200** to other neighboring cells.

Set of poly cut feature layout patterns **208** has a pattern width W_{1Y} (not labelled) in the second direction Y, and a pattern length L (not labelled) in the first direction X. In some embodiments, poly cut feature layout patterns **208a** and **208b** are usable to identify a corresponding location of a portion of corresponding gate structure **304a** and **304b** of integrated circuit **300** or **500** (FIG. 5) that is removed during operation **604** of method **600A** (FIG. 6A).

In some embodiments, the pattern width W_{1Y} (not labelled) corresponds to the cut width D_Y (not labelled) of one or more of gate structures **304a1**, **304a2**, **304b1** and **304b2**. In some embodiments, the pattern length L (not labelled) corresponds to the cut length L_Y (not labelled) of one or more of gate structures **304a1**, **304a2**, **304b1** and **304b2**. In some embodiments, at least one of the set of gate layout patterns **204**, the CPODE layout pattern **206** or the set of poly cut feature layout patterns **210** is located on a poly-gate layout level (POLY). Other configurations or quantities of patterns in the poly cut feature layout pattern **210** are within the scope of the present disclosure.

Layout design **200** further includes conductive feature layout patterns **210a**, **210b**, **210c**, **210d**, **210e** and **210f** (collectively referred to as “set of conductive feature layout patterns **210**”). In some embodiments, conductive feature layout patterns **210a**, **210b**, **210c**, **210d**, **210e** and **210f** are usable to manufacture corresponding conductive structures **310a**, **310b**, **310c**, **310d**, **310e** and **310f** of integrated circuit **300** (FIGS. 3A-3I) and **500** (FIG. 5).

In some embodiments, the set of conductive feature layout patterns **210** extends in the second direction Y, and is over the set of active region layout patterns **202**. Conductive feature layout patterns **210a** and **210b** overlap active region layout pattern **202a**. In some embodiments, conductive feature layout patterns **210a** and **210b** overlap cell boundary **250**. Conductive feature layout patterns **210c** and **210d** overlap active region layout pattern **202b**. In some embodiments, at least conductive feature layout pattern **210e** or **210f** overlaps active region layout pattern **202a**. In some embodiments, at least conductive feature layout pattern **210e** or **210f** overlaps active region layout pattern **202b**.

In some embodiments, at least conductive feature layout pattern **210e** or **210f** extends from active region layout pattern **202a** to active region layout pattern **202b**. In some embodiments, at least a side of conductive feature layout pattern **210e** or **210f** directly contacts or abuts a side of active region layout pattern **202b**. In some embodiments, at least conductive feature layout pattern **210e** or **210f** extends from active region layout pattern **202a** to a side of active region layout pattern **202b**.

In some embodiments, each conductive feature layout pattern of the set of conductive feature layout patterns **210** is separated from an adjacent layout pattern of the set of conductive feature layout patterns **210** in at least the first direction X or the second direction Y. The set of conductive feature layout patterns **210** is on a third layout level different from the first layout level and the second layout level. In some embodiments, the third layout level corresponds to the

metal over diffusion (MD) level of layout design **200** or **400** (FIG. 4). Other quantities or configurations of the set of conductive feature layout patterns **210** are within the scope of the present disclosure.

Layout design **200** further includes conductive feature layout patterns **212a**, **212b**, **212c**, **212d**, **212e** and **212f** (collectively referred to as “set of conductive feature layout patterns **212**”). In some embodiments, conductive feature layout patterns **212a**, **212b**, **212c**, **212d**, **212e** and **212f** are usable to manufacture corresponding conductive structures **312a**, **312b**, **312c**, **312d**, **312e** and **312f** of integrated circuit **300** (FIGS. 3A-3I) and **500** (FIG. 5).

In some embodiments, the set of conductive feature layout patterns **212** extends in the first direction X, and is over at least the set of active region layout patterns **210** or the set of gate layout patterns **204**.

Conductive feature layout pattern **212a** overlaps conductive feature layout patterns **210a** and **210b**, cut feature layout pattern **208a** and CPODE layout pattern **206**. Each of conductive feature layout patterns **212b** and **212c** overlaps conductive feature layout patterns **210e** and **210f**, and CPODE layout pattern **206**. Conductive feature layout patterns **212d** and **212e** overlap corresponding conductive feature layout patterns **210c** and **210d**. Conductive feature layout pattern **212b** overlaps conductive feature layout patterns **210a** and **210b**, cut feature layout pattern **208a** and CPODE layout pattern **206**.

Conductive feature layout pattern **212a** overlaps gate layout patterns **204a**, **204b** and **204c**. Conductive feature layout pattern **212b** overlaps gate layout patterns **204a** and **204c**. Conductive feature layout pattern **212c** overlaps gate layout patterns **204b** and **204c**. Conductive feature layout patterns **212d** and **212e** overlaps corresponding gate layout patterns **204a** and **204b**. Conductive feature layout pattern **212f** overlaps gate layout patterns **204a**, **204b** and **204c**, and CPODE layout pattern **206**.

In some embodiments, each conductive feature layout pattern of the set of conductive feature layout patterns **212** is separated from an adjacent layout pattern of the set of conductive feature layout patterns **212** in at least the first direction X or the second direction Y. The set of conductive feature layout patterns **212** is on a fourth layout level different from the first layout level, the second layout level and the third layout level. In some embodiments, the fourth layout level corresponds to the metal zero (M0) level of layout design **200** or **400** (FIG. 4). Other quantities or configurations of the set of conductive feature layout patterns **212** are within the scope of the present disclosure.

Layout design **200** further includes via layout patterns **214a**, **214b**, **214c**, **214d**, **214e**, **214f**, **214g** and **214h** (collectively referred to as “set of via layout patterns **214**”). In some embodiments, via layout patterns **214a**, **214b**, **214c**, **214d**, **214e**, **214f**, **214g** and **214h** are usable to manufacture corresponding vias **314a**, **314b**, **314c**, **314d**, **314e**, **314f**, **314g** and **314h** of integrated circuit **300** (FIGS. 3A-3I) and **500** (FIG. 5).

In some embodiments, via layout patterns **214c** and **214i** of the set of via layout patterns **214** are located where corresponding conductive feature layout patterns **212b** and **212f** of the set of conductive feature layout patterns **212** overlaps gate layout pattern **204a** of the set of gate layout patterns **204**.

In some embodiments, via layout patterns **214f** and **214j** of the set of via layout patterns **214** are located where corresponding conductive feature layout pattern **212c** and

212f of the set of conductive feature layout patterns **212** overlaps gate layout pattern **204b** of the set of gate layout patterns **204**.

In some embodiments, one or more of the set of via layout patterns **214** are between one or more of the set of conductive feature layout patterns **212** and one or more of the set of gate layout patterns **204**.

In some embodiments, via layout patterns **214a** and **214b** of the set of via layout patterns **214** are located where conductive feature layout pattern **212a** of the set of conductive feature layout patterns **212** overlaps corresponding conductive feature layout pattern **210a** and **210b** of the set of conductive feature layout patterns **210**.

In some embodiments, via layout patterns **214d** and **214e** of the set of via layout patterns **214** are located where corresponding conductive feature layout pattern **212b** and **212c** of the set of conductive feature layout patterns **212** overlaps corresponding conductive feature layout pattern **210f** and **210e** of the set of conductive feature layout patterns **210**.

In some embodiments, via layout patterns **214g** and **214h** of the set of via layout patterns **214** are located where corresponding conductive feature layout pattern **212d** and **212e** of the set of conductive feature layout patterns **212** overlaps corresponding conductive feature layout pattern **210c** and **210d** of the set of conductive feature layout patterns **210**.

In some embodiments, one or more of the set of via layout patterns **214** are between one or more of the set of conductive feature layout patterns **212** and one or more of the set of active region layout patterns **210**.

In some embodiments, via layout patterns **214c**, **214f**, **214i** and **214j** of the set of via layout patterns **214** are on at least the via over gate (VG) level of layout design **200** or **400** (FIG. 4). In some embodiments, the VG level is between the fourth layout level and the second layout level of layout design **200** or **400** (FIG. 4).

In some embodiments, via layout patterns **214a**, **214b**, **214d**, **214e**, **214g** and **214h** of the set of via layout patterns **214** are on at least the via over diffusion (VD) level of layout design **200** or **400** (FIG. 4). In some embodiments, the VD level is between the fourth layout level and the third layout level of layout design **200** or **400** (FIG. 4).

Other quantities or configurations of the set of via layout patterns **214** are within the scope of the present disclosure.

Layout design **200** further includes conductive feature layout patterns **218a**, **218b**, **218c**, **218d** and **218e** (collectively referred to as “set of conductive feature layout patterns **218**”) as shown in FIG. 2B. In some embodiments, conductive feature layout patterns **218a**, **218b**, **218c**, **218d** and **218e** are usable to manufacture corresponding conductive structures **318a**, **318b**, **318c**, **318d** and **318e** of integrated circuit **300** (FIGS. 3A-3I) and **500** (FIG. 5).

In some embodiments, the set of conductive feature layout patterns **218** extends in the second direction Y, and is over at least the set of active region layout patterns **210**, the set of gate layout patterns **204**, or the set of conductive feature layout patterns **212**.

Conductive feature layout pattern **218a** overlaps active region layout patterns **202a** and **202b**, cut feature layout pattern **208b**, conductive feature layout pattern **212d** and at least a portion of conductive feature layout patterns **210a** and **210c**.

Conductive feature layout pattern **218b** overlaps active region layout patterns **202a** and **202b**, cut feature layout patterns **208a** and **208b**, conductive feature layout patterns

212a, **212b**, **212c**, **212d** and **212f**, gate layout pattern **204a**, and at least a portion of conductive feature layout pattern **210e**.

Conductive feature layout pattern **218c** overlaps active region layout patterns **202a** and **202b**, cut feature layout patterns **208a** and **208b**, CPODE layout pattern **206**, conductive feature layout patterns **212a**, **212b**, **212c** and **212f**, and gate layout pattern **204b**.

Conductive feature layout pattern **218d** overlaps active region layout patterns **202a** and **202b**, cut feature layout patterns **208a** and **208b**, conductive feature layout patterns **212a**, **212b**, **212c**, **212e** and **212f**, gate layout pattern **204c**, and at least a portion of conductive feature layout pattern **210f**.

Conductive feature layout pattern **218e** overlaps active region layout patterns **202a** and **202b**, cut feature layout pattern **208b**, conductive feature layout pattern **212e** and at least a portion of conductive feature layout patterns **210b** and **210d**.

In some embodiments, each conductive feature layout pattern of the set of conductive feature layout patterns **218** is separated from an adjacent layout pattern of the set of conductive feature layout patterns **218** in at least the first direction X. The set of conductive feature layout patterns **218** is on a fifth layout level different from the first layout level, the second layout level, the third layout level and the fourth layout level. In some embodiments, the fifth layout level corresponds to the metal one (M1) level of layout design **200** or **400** (FIG. 4). Other quantities or configurations of the set of conductive feature layout patterns **218** are within the scope of the present disclosure.

Layout design **200** further includes via layout patterns **220a**, **220b**, **220c**, **220d** and **220e** (collectively referred to as “set of via layout patterns **220**”). In some embodiments, via layout patterns **220a**, **220b**, **220c**, **220d** and **220e** are usable to manufacture corresponding vias **320a**, **320b**, **320c**, **320d** and **320e** of integrated circuit **300** (FIGS. 3A-3I) and **500** (FIG. 5).

In some embodiments, one or more of the set of via layout patterns **220** are between one or more of the set of conductive feature layout patterns **218** and one or more of the conductive feature layout patterns **212**.

In some embodiments, via layout patterns **220a** and **220e** of the set of via layout patterns **220** are located where corresponding conductive feature layout patterns **218a** and **218e** of the set of conductive feature layout patterns **218** overlap corresponding conductive feature layout patterns **212b** and **212e** of the set of conductive feature layout patterns **212**.

In some embodiments, via layout patterns **220b** and **220c** of the set of via layout patterns **220** are located where corresponding conductive feature layout patterns **218b** and **218d** of the set of conductive feature layout patterns **218** overlap conductive feature layout patterns **212f** of the set of conductive feature layout patterns **212**.

In some embodiments, via layout pattern **220d** of the set of via layout patterns **220** is located where conductive feature layout pattern **218c** of the set of conductive feature layout patterns **218** overlaps conductive feature layout pattern **212a** of the set of conductive feature layout patterns **212**.

In some embodiments, the set of via layout patterns **220** are on at least the via zero (V0) level of layout design **200** or **400** (FIG. 4). In some embodiments, the V0 level is between the fourth layout level and the fifth layout level of layout design **200** or **400** (FIG. 4). Other quantities or

configurations of the set of via layout patterns **220** are within the scope of the present disclosure.

Layout design **200** further includes conductive feature layout patterns **222a**, **222b**, **222c**, **222d** and **222e** (collectively referred to as “set of conductive feature layout patterns **222**”). In some embodiments, conductive feature layout patterns **222a**, **222b**, **222c**, **222d** and **222e** are usable to manufacture corresponding conductive structures **322a**, **322b**, **322c**, **322d** and **322e** of integrated circuit **300** (FIGS. 3A-3I) and **500** (FIG. 5).

In some embodiments, the set of conductive feature layout patterns **222** extends in the first direction X, and is over at least the set of active region layout patterns **210**, the set of gate layout patterns **204**, the set of conductive feature layout patterns **212** or the set of conductive feature layout patterns **218**.

Conductive feature layout pattern **222a** overlaps cut feature layout pattern **208a**, conductive feature layout pattern **212a**, conductive feature layout patterns **210a** and **210b**, gate layout patterns **204a**, **204b** and **204c**, CPODE layout pattern **206** and conductive feature layout patterns **218b**, **218c** and **218d**.

Conductive feature layout pattern **222b** overlaps active region layout pattern **202a**, conductive feature layout patterns **212b**, gate layout patterns **204a**, **204b** and **204c**, CPODE layout pattern **206**, conductive feature layout patterns **218a**, **218b**, **218c**, **218d** and **218e**, and at least a portion of conductive feature layout patterns **210a**, **210b**, **210e** and **210f**.

Conductive feature layout pattern **222c** overlaps conductive feature layout patterns **212c**, gate layout patterns **204a**, **204b** and **204c**, CPODE layout pattern **206**, conductive feature layout patterns **218a**, **218b**, **218c**, **218d** and **218e**, and at least a portion of conductive feature layout patterns **210e** and **210f**.

Conductive feature layout pattern **222d** overlaps conductive feature layout patterns **212d** and **212e**, gate layout patterns **204a**, **204b** and **204c**, CPODE layout pattern **206**, conductive feature layout patterns **218a**, **218b**, **218c**, **218d** and **218e**, and at least a portion of conductive feature layout patterns **210c**, **210d**, **210e** and **210f**. In some embodiments, conductive feature layout pattern **222d** overlaps at least a portion of active region layout pattern **202b**.

Conductive feature layout pattern **222e** overlaps conductive feature layout pattern **212f**, gate layout patterns **204a**, **204b** and **204c**, CPODE layout pattern **206** and conductive feature layout patterns **218b**, **218c** and **218d**.

In some embodiments, each conductive feature layout pattern of the set of conductive feature layout patterns **222** is separated from an adjacent layout pattern of the set of conductive feature layout patterns **222** in at least the second direction Y. The set of conductive feature layout patterns **222** is on a sixth layout level different from the first layout level, the second layout level, the third layout level, the fourth layout level and the fifth layout level. In some embodiments, the sixth layout level corresponds to the metal two (M2) level of layout design **200** or **400** (FIG. 4). Other quantities or configurations of the set of conductive feature layout patterns **222** are within the scope of the present disclosure.

Layout design **200** further includes via layout patterns **226a**, **226b**, **226c**, **226d** and **226e** (collectively referred to as “set of via layout patterns **226**”). In some embodiments, via layout patterns **226a**, **226b**, **226c**, **226d** and **226e** are usable to manufacture corresponding vias **326a**, **326b**, **326c**, **326d** and **326e** of integrated circuit **300** (FIGS. 3A-3I) and **500** (FIG. 5).

In some embodiments, one or more of the set of via layout patterns **226** are between one or more of the set of conductive feature layout patterns **222** and one or more of the conductive feature layout patterns **218**.

In some embodiments, via layout pattern **226a** of the set of via layout patterns **226** is located where corresponding conductive feature layout pattern **222b** of the set of conductive feature layout patterns **222** overlaps conductive structure **218a** of the set of conductive feature layout patterns **218**.

In some embodiments, via layout patterns **226b**, **226c** and **226d** of the set of via layout patterns **226** are located where corresponding conductive feature layout patterns **222e**, **222c** and **222a** of the set of conductive feature layout patterns **222** overlap conductive structure **218c** of the set of conductive feature layout patterns **218**.

In some embodiments, via layout pattern **226e** of the set of via layout patterns **226** is located where corresponding conductive feature layout pattern **222d** of the set of conductive feature layout patterns **222** overlaps conductive structure **218e** of the set of conductive feature layout patterns **218**.

In some embodiments, each center of via layout patterns **220d** and **220e** is aligned with a corresponding center of corresponding via layout patterns **226d** and **226e**.

In some embodiments, at least a center of via layout pattern **220e**, **226e** or **214h** is aligned with at least another center of via layout pattern **220e**, **226e** or **214h**.

In some embodiments, a center of via layout pattern **220a** is aligned with a center of via layout pattern **214g**.

In some embodiments, the set of via layout patterns **226** are on at least the via one (V1) level of layout design **200** or **400** (FIG. 4). In some embodiments, the V1 level is between the fifth layout level and the sixth layout level of layout design **200** or **400** (FIG. 4). Other quantities or configurations of the set of via layout patterns **226** are within the scope of the present disclosure.

In some embodiments, by using conductive feature layout pattern **222b** and **222d** between conductive feature layout patterns **222a**, **222c**, **222e** of layout design **200** results in layout design **200** or **400** having better electromagnetic shielding than other approaches. In some embodiments, by providing better electromagnetic shielding results in layout design **200** or **400** having less cross talk than other approaches.

Integrated Circuit—Cross-Sectional Views

FIGS. 3A, 3B, 3C, 3D, 3E, 3F, 3G, 3H and 3I are diagrams of an integrated circuit **300**, in accordance with some embodiments.

FIG. 3A is a cross-sectional view of an integrated circuit **300** corresponding to layout design **200** as intersected by plane A-A'. FIG. 3B is a cross-sectional view of an integrated circuit **300** corresponding to layout design **200** as intersected by plane B-B'. FIG. 3C is a cross-sectional view of an integrated circuit **300** corresponding to layout design **200** as intersected by plane C-C'. FIG. 3D is a cross-sectional view of an integrated circuit **300** corresponding to layout design **200** as intersected by plane D-D'. FIG. 3E is a cross-sectional view of an integrated circuit **300** corresponding to layout design **200** as intersected by plane E-E'. FIG. 3F is a cross-sectional view of an integrated circuit **300** corresponding to layout design **200** as intersected by plane F-F'. FIG. 3G is a cross-sectional view of an integrated circuit **300** corresponding to layout design **200** as intersected by plane G-G'. FIG. 3H is a cross-sectional view of an integrated circuit **300** corresponding to layout design **200** as intersected by plane H-H'. FIG. 3I is a cross-sectional view

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of an integrated circuit **300** corresponding to layout design **200** as intersected by plane I-I'.

Integrated circuit **300** is manufactured by layout design **200**. Integrated circuit **300** is a structural implementation of memory cell **100B** or **100C**.

Structural relationships including alignment, lengths and widths, as well as configurations of integrated circuit **300** or **500** are similar to the structural relationships and configurations of layout design **200** of FIGS. 2A-2B and layout design **400** of FIGS. 4A-4B, and will not be described in FIGS. 3A-3I for brevity.

Integrated circuit **300** includes a first well **301** and a second well **302**. Each of the first well **301** and the second well **302** is located on at least the first level of integrated circuit **300**, and extends in at least the first direction X or the second direction Y. In some embodiments, the first well **301** and the second well **302** are adjacent to each other, and are separated from each other in the second direction Y.

The first well **301** includes a first portion **301a** and a second portion **301b**.

The second well **302** includes a first portion **302a** and a second portion **302b**.

The first portion **301a** and the second portion **301b** of the first well **301** of integrated circuit **300** include dopants of a first type. The first portion **302a** and the second portion **302b** of the second well **302** of integrated circuit **300** include dopants of a second type different from the first type.

In some embodiments, the first type is an N-type dopant and the second type is a P-type dopant. In some embodiments, the first type is a P-type dopant and the second type is an N-type dopant.

The first portion **301a** and the second portion **301b** of the first well **301** of integrated circuit **300** extend in the first direction X. In some embodiments, the first portion **301a** and the second portion **301b** are a continuous part of the first well **301**. In some embodiments, the first portion **301a** and the second portion **301b** are separated by an insulating portion **330**.

The first portion **302a** and the second portion **302b** of the second well **302** of integrated circuit **300** extend in the first direction X. In some embodiments, the first portion **302a** and the second portion **302b** are a continuous part of the second well **302**. In some embodiments, the first portion **302a** and the second portion **302b** are separated by the insulating portion **330**. In some embodiments, the insulating portion **330** extends in the second direction Y.

Other quantities or configurations of the first well **301** or the second well **302** are within the scope of the present disclosure.

Integrated circuit **300** further includes a set of active regions **303** and a set of active regions **305**. The set of active regions **303** and the set of active regions **305** extend in the second direction Y. The set of active regions **303** and the set of active regions **305** are located on the first level of integrated circuit **300**.

The set of active regions **303** includes one or more of active regions **303a1**, **303a2**, **303b1** and **303b2**. Each of the active regions **303a1**, **303a2**, **303b1** and **303b2** of the set of active regions **303** is separated from an adjacent active region of the set of active regions **303** in the first direction X by a first pitch (not labelled).

Active regions **303a1** and **303a2** of the set of active regions **303** are embedded in the first portion **301a** of the second well **302** of integrated circuit **300**. Active region **303b1** and **303b2** of the set of active regions **303** are embedded in the second portion **301b** of the second well **302**

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of integrated circuit **300**. Active regions **303a1**, **303a2**, **303b1** and **303b2** include dopants of the first type.

The set of active regions **305** includes one or more of active regions **305a1**, **305a2**, **305b1** and **305b2**. Each of the active regions **305a1**, **305a2**, **305b1** and **305b2** of the set of active regions **305** is separated from an adjacent active region of the set of active regions **305** in the first direction X by a second pitch (not labelled).

Active regions **305a1** and **305a2** of the set of active regions **305** are embedded in the first portion **301a** of the first well **301** of integrated circuit **300**. Active region **305b1** and **305b2** of the set of active regions **305** are embedded in the second portion **301b** of the first well **301** of integrated circuit **300**. Active regions **305a1**, **305a2**, **305b1** and **305b2** include dopants of the second type.

In some embodiments, integrated circuit **300** corresponds to memory cell **100B** of FIG. 1B, and the first type is an N-type dopant and the second type is a P-type dopant, such that the first well **301** of integrated circuit **300** is an N-well, the second well **302** of integrated circuit **300** is a P-well, active regions **303a1**, **303a2**, **303b1** and **303b2** are n-type dopants embedded in the second well **302** (P-well), and active regions **305a1**, **305a2**, **305b1** and **305b2** are p-type dopants embedded in the first well **301** (N-well). In these embodiments, the first well **301** corresponds to the N-well of PMOS transistors PG0 and PG1 of memory cell **100B** of FIG. 1B, and the second well **302** corresponds to the P-well of NMOS transistors PD0 and PD1 of memory cell **100B** of FIG. 1B. In these embodiments, active regions **303a1** and **303b2** are corresponding source regions of corresponding NMOS transistors PD0 and PD1 of memory cell **100B** of FIG. 1B, and active regions **303a2** and **303b1** are corresponding drain regions of corresponding NMOS transistors PD0 and PD1 of memory cell **100B** of FIG. 1B. In these embodiments, active regions **305a1** and **305b2** are corresponding source or drain regions of corresponding PMOS transistors PG0 and PG1 of memory cell **100B** of FIG. 1B, and active regions **305a2** and **305b1** are corresponding drain or source regions of corresponding PMOS transistors PG0 and PG1 of memory cell **100B** of FIG. 1B.

In some embodiments, integrated circuit **300** corresponds to memory cell **100C** of FIG. 1C, and the first type is a P-type dopant and the second type is an N-type dopant, such that the first well **301** of integrated circuit **300** is a P-well, and the second well **302** of integrated circuit **300** is an N-well, active regions **303a1**, **303a2**, **303b1** and **303b2** are p-type dopants embedded in the second well **302** (N-well), and active regions **305a1**, **305a2**, **305b1** and **305b2** are n-type dopants embedded in the first well **301** (P-well). In these embodiments, integrated circuit **300** corresponds to memory cell **100C** of FIG. 1C, and the first well **301** corresponds to the P-well of NMOS transistors PG2 and PG3, and the second well **302** corresponds to the N-well of PMOS transistors PU0 and PU1 of memory cell **100C** of FIG. 1C. In these embodiments, active regions **303a1** and **303b2** are corresponding source regions of corresponding PMOS transistors PU0 and PU1 of memory cell **100C** of FIG. 1C, and active regions **303a2** and **303b1** are corresponding drain regions of corresponding PMOS transistors PU0 and PU1 of memory cell **100C** of FIG. 1C. In these embodiments, active regions **305a1** and **305b2** are corresponding source or drain regions of corresponding NMOS transistors PG2 and PG3 of memory cell **100C** of FIG. 1C, and active regions **305a2** and **305b1** are corresponding drain or source regions of corresponding NMOS transistors PG2 and PG3 of memory cell **100C** of FIG. 1C.

Other quantities or configurations of the set of active regions **303** or **305** are within the scope of the present disclosure.

Integrated circuit **300** further includes a set of gates **304** extending in the second direction Y. The set of gates **304** are over the set of active regions **302** and **303** of integrated circuit **300**. The set of gates **304** is located on a second level of integrated circuit **300** or **500**. The second level is above the first level of integrated circuit **300** or **500**. In some embodiments, the second level of integrated circuit **300** or **500** is referred to as the POLY level.

The set of gates **304** includes one or more of gate structures **304a** or **304b**. Gate structures **304a** and **304b** are separated from each other in the first direction X. Insulating portion **330** is between gate structures **304a** and **304b**, and is separated from each gate structure **304a** or **304b** by a gate pitch (not labelled).

Gate structure **304a** includes gate structures **304a1** and **304a2**. Gate structures **304a1** and **304a2** are separated from each other in the second direction Y by the cut width D_V (not labelled).

Gate structure **304b** includes gate structures **304b1** and **304b2**. Gate structures **304b1** and **304b2** are separated from each other in the second direction Y by the cut width D_V (not labelled).

In some embodiments, integrated circuit **300** corresponds to memory cell **100B** of FIG. 1B, such that gate structures **304a1** and **304b1** are the corresponding gates of NMOS transistors PD0 and PD1, and gate structures **304a2** and **304b2** are the corresponding gates of PMOS transistors PG0 and PG1. In some embodiments, integrated circuit **300** corresponds to memory cell **100C** of FIG. 1C, such that gate structures **304a1** and **304b1** are the corresponding gates of PMOS transistors PU0 and PU1, and gate structures **304a2** and **304b2** are the corresponding gates of NMOS transistors PG2 and PG3.

Other quantities or configurations of at least the set of gates **304**, gate structures **304a1**, **304a2**, **304b1** or **304b2** are within the scope of the present disclosure.

Integrated circuit **300** further includes conductive structures **310b**, **310c**, **310d**, **310e** and **310f** (collectively referred to as a “set of contacts **310**”). Set of contacts **310** extends in the second direction Y. The set of contacts **310** is over the set of active regions **303**, the set of active regions **305**, the first well **301** and the second well **302**. The set of contacts **310** is located on the second level of integrated circuit **300** or **500**. In some embodiments, the second level of integrated circuit **300** or **500** is referred to as the metal diffusion (MD) level.

The set of contacts **310** is electrically coupled to the set of active regions **303** or **305** of integrated circuit **300** or **500**. In some embodiments, the set of contacts **310** electrically couples the set of active regions **303** or **305** of integrated circuit **300** or **500** to upper levels (e.g., M0, M1 or M2) of integrated circuit **300** or **500**.

Conductive structures **310a**, **310b**, **310c**, **310d** electrically couple corresponding vias **314a**, **314b**, **314g**, **314h** to corresponding active regions **303a1**, **303b2**, **305a1**, **305b2**. Conductive structures **310e**, **310f** electrically couple corresponding active regions **303a2**, **303b1** to corresponding active regions **305a2**, **305b1**. Conductive structures **310e**, **310f** electrically couple corresponding vias **314e**, **314d** to corresponding conductive structures **312c**, **312b** (M0 level). Other quantities or configurations of the set of contacts **310** are within the scope of the present disclosure.

Integrated circuit **300** further includes conductive structures **312a**, **312b**, **312c**, **312d**, **312e** and **312f** (collectively

referred to as a “set of conductive structures **312**”). Set of conductive structures **312** extends in the first direction X. The set of conductive structures **312** is over the set of contacts **310**, the set of gate structures **304**, the set of active regions **303**, the set of active regions **305**, the first well **301** and the second well **302** of integrated circuit **300** or **500**. The set of conductive structures **312** is located on a third level of integrated circuit **300** or **500**. The third level of integrated circuit **300** or **500** is above the first level and the second level of integrated circuit **300** or **500**. In some embodiments, the third level of integrated circuit **300** or **500** is referred to as the metal zero (M0) level.

The set of conductive structures **312** electrically couples the set of active regions **303** or **305** to upper levels (e.g., M1 or M2) of integrated circuit **300** or **500**. In some embodiments, the set of conductive structures **312** electrically couples the set of gates **304** to upper levels (e.g., M1 or M2) of integrated circuit **300** or **500**.

Conductive structure **312a** electrically couples conductive structure **310a** to conductive structure **310b** by vias **314a** and **314b**. Conductive structure **312b** electrically couples conductive structure **310f** to gate structure **304a1** by vias **314c** and **314d**. Conductive structure **312c** electrically couples conductive structure **310e** to gate structure **304b1** by vias **314e** and **314f**.

Conductive structure **312d** electrically couples conductive structure **310c** to upper layers (e.g., conductive structure **318a**) by vias **314g** and **320a**. Conductive structure **312e** electrically couples conductive structure **310d** to upper layers (e.g., conductive structure **318e**) by vias **314h** and **320e**.

Conductive structure **312f** electrically couples gate structure **304a2** to gate structure **304b2** by vias **314i** and **314j**. Other quantities or configurations of the set of conductive structures **312** are within the scope of the present disclosure.

Integrated circuit **300** further includes vias **314a**, **314b**, **314c**, **314d**, **314e**, **314f**, **314g**, **314h**, **314i** and **314j** (collectively referred to as a “set of vias **314**”) between the set of conductive structures **312** and either the set of conductive structures **310** or the set of gate structures **304** of integrated circuit **300** or **500**. The set of vias **314** electrically couple the set of conductive structures **312** to the set of conductive structures **310** or the set of gate structures **304** of integrated circuit **300** or **500**.

In some embodiments, one or more vias of the set of vias **314** is located where one or more conductive structures of the set of conductive structures **312** is over one or more of the set of conductive structures **310** or the set of gate structures **304** of integrated circuit **300** or **500**.

Vias **314a**, **314b** electrically couple conductive structure **312a** to corresponding conductive structures **310a**, **310b**. Vias **314c**, **314f** electrically couple corresponding conductive structures **312b**, **312c** to corresponding gate structures **304a2**, **304b2**. Vias **314d**, **314e** electrically couple corresponding conductive structures **312b**, **312c** to corresponding conductive structures **310f**, **310e**. Vias **314g**, **314h** electrically couples corresponding conductive structures **312d**, **312e** to corresponding conductive structures **310c**, **310d**.

Set of vias **314** is in the via over diffusion (VD) level or the via over gate (VG) level of integrated circuit **300** or **500**. The VG or VD level of integrated circuit **300** or **500** is between the second level and the third level. In some embodiments, vias **314c**, **314f**, **314i** and **314j** are in the VG level of integrated circuit **300** or **500**. In some embodiments, vias **314a**, **314b**, **314d**, **314e**, **314g** and **314h** are in the VD

level of integrated circuit 300 or 500. Other quantities or configurations of the set of vias 314 are within the scope of the present disclosure.

Integrated circuit 300 further includes conductive structures 318a, 318b, 318c, 318d and 318e (collectively referred to as a “set of conductive structures 318”). Set of conductive structures 318 extends in the second direction Y. The set of conductive structures 318 is over the set of vias 320, set of conductive structures 312, the set of contacts 310, the set of gate structures 304, the set of active regions 303, the set of active regions 305, the first well 301 and the second well 302 of integrated circuit 300 or 500. The set of conductive structures 318 is located on a fourth level of integrated circuit 300 or 500. The fourth level of integrated circuit 300 or 500 is above the first level, the second level and the third level of integrated circuit 300 or 500. In some embodiments, the fourth level of integrated circuit 300 or 500 is referred to as the metal zero (M0) level.

The set of conductive structures 318 electrically couples the set of conductive structures 312 to upper levels (e.g., the set of conductive structures 322 (M1)) of integrated circuit 300 or 500.

Conductive structure 318a corresponds to at least a portion of the bit line BL of memory cell 100B of FIG. 1B or of memory cell 100C of FIG. 1C.

Together, conductive structures 318b and 318d correspond to the word line WL of memory cell 100B of FIG. 1B or memory cell 100C of FIG. 1C.

Conductive structure 318e corresponds to at least a portion of the bit line bar BLB of memory cell 100B of FIG. 1B or of memory cell 100C of FIG. 1C.

In some embodiments, integrated circuit 300 corresponds to memory cell 100B of FIG. 1B, and conductive structure 318c is electrically coupled to a reference supply voltage VSS. In some embodiments, integrated circuit 300 corresponds to memory cell 100C of FIG. 1C, and conductive structure 318c is electrically coupled to a supply voltage VDD. Other quantities or configurations of the set of conductive structures 318 are within the scope of the present disclosure.

Integrated circuit 300 further includes vias 320a, 320b, 320c, 320d and 320e (collectively referred to as a “set of vias 320”) between the set of conductive structures 318 and the set of conductive structures 312 of integrated circuit 300 or 500. The set of vias 320 electrically couple the set of conductive structures 318 to the set of conductive structures 312 of integrated circuit 300 or 500. In some embodiments, one or more vias of the set of vias 320 is located where one or more conductive structures of the set of conductive structures 318 is over one or more of the set of conductive structures 312 of integrated circuit 300 or 500.

Vias 320a, 320e electrically couple corresponding conductive structures 318a, 318e to corresponding conductive structures 312d, 312e. Vias 320b, 320c electrically couple corresponding conductive structures 318b, 318d to corresponding conductive structures 312f. Via 320d electrically couples conductive structure 318c to conductive structure 312a.

Set of vias 320 is in the via zero (V0) level of integrated circuit 300 or 500. The V0 level of integrated circuit 300 or 500 is between the third level and the fourth level. Other quantities or configurations of the set of vias 320 are within the scope of the present disclosure.

Integrated circuit 300 further includes conductive structures 322a, 322b, 322c, 322d and 322e (collectively referred to as a “set of conductive structures 322”). Set of conductive structures 322 extends in the first direction X. The set of

conductive structures 322 is over the set of vias 326, set of vias 320, set of conductive structures 318, set of conductive structures 312, the set of contacts 310, the set of gate structures 304, the set of active regions 303, the set of active regions 305, the first well 301 and the second well 302 of integrated circuit 300 or 500. The set of conductive structures 322 is located on a fifth level of integrated circuit 300 or 500. The fifth level of integrated circuit 300 or 500 is above the first level, the second level, the third level and the fourth level of integrated circuit 300 or 500. In some embodiments, the fifth level of integrated circuit 300 or 500 is referred to as the metal one (M1) level.

The set of conductive structures 322 is electrically coupled to the set of conductive structures 312 and the other lower levels (e.g., M0, Poly, OD, etc.) of integrated circuit 300 or 500. In some embodiments, the set of conductive structures 322 is electrically coupled to other upper levels (not shown) of integrated circuit 300 or 500.

Conductive structure 322b and 318a corresponds to at least a portion of the bit line BL of memory cell 100B of FIG. 1B or of memory cell 100C of FIG. 1C. Conductive structure 322d and 318e corresponds to at least a portion of the bit line bar BLB of memory cell 100B of FIG. 1B or of memory cell 100C of FIG. 1C.

In some embodiments, integrated circuit 300 corresponds to memory cell 100B of FIG. 1B, and conductive structures 322a, 322c, 322e and 318c are electrically coupled to the reference supply voltage VSS. In some embodiments, integrated circuit 300 corresponds to memory cell 100C of FIG. 1C, and conductive structures 322a, 322c, 322e and 318c are electrically coupled to the supply voltage VDD. Other quantities or configurations of the set of conductive structures 322 are within the scope of the present disclosure.

Integrated circuit 300 further includes vias 326a, 326b, 326c, 326d and 326e (collectively referred to as a “set of vias 326”) between the set of conductive structures 322 and the set of conductive structures 318 of integrated circuit 300 or 500. The set of vias 326 electrically couple the set of conductive structures 322 to the set of conductive structures 318 of integrated circuit 300 or 500. In some embodiments, one or more vias of the set of vias 326 is located where one or more conductive structures of the set of conductive structures 322 is over one or more of the set of conductive structures 318 of integrated circuit 300 or 500.

Vias 326a, 326e electrically couple corresponding conductive structures 322b, 322d to corresponding conductive structures 318a, 318e. Vias 326b, 326c, 326d electrically couple corresponding conductive structures 322e, 322c, 322a to conductive structure 318c.

Set of vias 326 is in the via one (V1) level of integrated circuit 300 or 500. The V1 level of integrated circuit 300 or 500 is between the fourth level and the fifth level. Other quantities or configurations of the set of vias 326 are within the scope of the present disclosure.

In some embodiments, at least one conductive structure of the set of contacts 310 or at least one conductive structure of the set of conductive structures 312, 318 or 322 is a conductive material including copper, aluminum, alloys thereof or other suitable conductive materials, that is formed in one or more metallization layers by one or more of a physical vapor deposition process, a chemical vapor deposition process, a plating process, or other suitable processes.

In some embodiments, at least one via of the set of vias 314, 320 or 326 is a metal line, a via, a through silicon via (TSV), an inter-level via (ILV), a slot via, an array of vias, or another suitable conductive line. In some embodiments, at least one via of the set of vias 314, 320 or 326 includes

copper, aluminum, nickel, titanium, tungsten, cobalt, carbon, alloys thereof or another suitable conductive material, that is formed in one or more metallization layers by one or more of a physical vapor deposition process, a chemical vapor deposition process, a plating process, or other suitable processes. In some embodiments, at least one via of the set of vias **314**, **320** or **326** includes one or more conductive line segments. Other configurations, materials or quantities of the set of vias **314**, **320** or **326** are within the scope of the present disclosure.

In some embodiments, integrated circuit **300** or **500** (FIG. **5**) occupies less area than other integrated circuits. In some embodiments, by occupying less area than other integrated circuits, integrated circuit **300** or **500** is utilized as part of a memory macro **100A** that is denser compared with other approaches. In some embodiments, by being utilized as part of a denser memory macro **100A** or memory cell array, memory macro **100A** has a larger memory capacity than other approaches.

In some embodiments, by using conductive structures **322b** and **322d** between conductive structures **322a**, **322c**, **322e** results in integrated circuit **300** having better electromagnetic shielding than other approaches. In some embodiments, by providing better electromagnetic shielding results in integrated circuit **300** having less cross talk than other approaches.

Layout Design

FIGS. **4A-4B** are diagrams of a layout design **400**, in accordance with some embodiments.

Layout design **400** is a variation of layout design **200** of FIGS. **2A-2C**. For example, layout design **400** has bit line layout patterns and bit line bar layout patterns with a greater width than layout design **200**. Components that are the same or similar to those in one or more of FIGS. **2A-2C** are given the same reference numbers, and detailed description thereof is thus omitted.

Layout design **400** is a layout diagram of memory cell **100B** of FIG. **1B** or memory cell **100C** of FIG. **1C**. Layout design **400** is usable to manufacture memory cell **100B** or **100C**. Layout design **400** includes portion **200A** (FIG. **2A**) and a portion **400A** (FIG. **4A**). For case of illustration, layout design **400** of FIG. **4A** does not include portion **200A**. Layout design **400**, as shown in FIG. **4B**, includes portion **200A** of FIG. **2A** and portion **400A** of FIG. **4A**.

In comparison with layout design **200** of FIGS. **2A-2C**, layout design **400** does not include conductive feature layout patterns **222a** and **222e**, and via layout patterns **226b** and **226d**.

In comparison with layout design **200** of FIGS. **2A-2C**, conductive feature layout pattern **422d** of layout design **400** replaces conductive feature layout pattern **222a**, and conductive feature layout pattern **422b** of layout design **400** replaces conductive feature layout pattern **222b**.

Conductive feature layout patterns **422d** and **422b** are similar to corresponding conductive feature layout patterns **222d** and **222b**, and similar detailed description is therefore omitted. In some embodiments, conductive feature layout patterns **422d** and **422b** are usable to manufacture corresponding conductive structures **522d** and **522b** (FIG. **5**).

Conductive feature layout patterns **422d** or **422b** has a larger width in the second direction **Y** than a width of corresponding conductive feature layout pattern **222d** or **222b** in the second direction **Y**. By increasing the width of conductive feature layout patterns **422d**, **422b**, conductive feature layout patterns **422d**, **422b** are usable to manufacture corresponding bit lines **BL** (e.g., conductive structure **522b** in FIG. **5**) and bit line bars **BLB** (e.g., conductive structure

522d in FIG. **5**) having less resistance than approaches with a smaller width in the second direction **Y**. In some embodiments, by removing conductive feature layout patterns **222a** and **222e**, and via layout patterns **226b** and **226d**, layout design **400** has increased routing resources usable for other metal layers.

Other configurations of conductive feature layout patterns **422d** or **422b** are within the scope of the present disclosure. For example, in some embodiments, the width or length of at least conductive feature layout patterns **422d** or **422b** is changed to overlap or expose different layout patterns than that shown in FIG. **4**.

Integrated Circuit—Cross-Sectional Views

FIG. **5** is a cross-sectional view of an integrated circuit **500** corresponding to layout design **400** as intersected by plane J-J'. FIG. **5** is a variation of FIG. **3H** of integrated circuit **300**.

Integrated circuit **500** is manufactured by layout design **400**. Integrated circuit **500** is a structural implementation of memory cell **100B** or **100C**.

Integrated circuit **500** is a variation of integrated circuit **300**. In comparison with integrated circuit **300**, integrated circuit **500** does not include conductive structures **322a** and **322e**, and vias **326b** and **326d**.

In comparison with integrated circuit **300** of FIGS. **3A-3I**, conductive structure **522b** of integrated circuit **500** replaces conductive structure **322b**, and conductive structure **522d** of integrated circuit **500** replaces conductive structure **322d**.

Conductive structures **522b** and **522d** are similar to corresponding conductive structures **322b** and **322d**, and similar detailed description is therefore omitted.

Conductive structures **522b** or **522d** has a larger width in the second direction **Y** than a width of corresponding conductive structures **322b** or **322d** in the second direction **Y**. By increasing the width of conductive structures **522b** and **522d**, bit line **BL** (e.g., conductive structure **522b** in FIG. **5**) and bit line bar **BLB** (e.g., conductive structure **522d** in FIG. **5**) have less resistance than approaches with a smaller width in the second direction **Y**. In some embodiments, by removing conductive structures **322a** and **322e**, and vias **326b** and **326d**, integrated circuit **500** has increased routing resources usable for other metal layers.

Other configurations of conductive structures **522b** or **522d** are within the scope of the present disclosure. For example, in some embodiments, the width or length of at least conductive structure **522b** or **522d** is changed to overlap or expose different portions of integrated circuit **500** than that shown in FIG. **5**.

For brevity, a single cross-sectional figure (e.g., FIG. **5**) of integrated circuit **500** is shown. However, it is understood that integrated circuit **500** includes additional cross-sectional figures (not shown) that are similar to the cross-sectional figures of integrated circuit **300** shown in FIGS. **3A-3I**. For example, each of the cross-sectional views of integrated circuit **300** can be modified, similar to FIG. **5**, to not include conductive structures **322a** and **322e**, and vias **326b** and **326d**, and conductive structure **522b** replaces conductive structure **322b**, and conductive structure **522d** replaces conductive structure **322d**.

Method

FIG. **6A** is a flowchart of a method **600A** of forming or manufacturing an IC in accordance with some embodiments. It is understood that additional operations may be performed before, during, and/or after the method **600A** depicted in FIG. **6A**, and that some other processes may only be briefly

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described herein. In some embodiments, the method **600A** is usable to form integrated circuits, such as memory macro **100A** (FIG. 1A), memory cell **100B** (FIG. 1B), memory cell **100C** (FIG. 1C), IC structure **300** (FIGS. 3A-3I) or IC structure **500** (FIG. 5). In some embodiments, the method **600A** is usable to form integrated circuits having similar relationships as one or more of layout designs **200** or **400** (FIGS. 2A-2C or 4A-4B).

In operation **602** of method **600A**, a layout design **200** or **400** of a memory array circuit (e.g., memory cell **100B**, **100C**) is generated. Operation **602** is performed by a processing device (e.g., processor **702** (FIG. 7)) configured to execute instructions for generating a layout design **200**, **400** or **716** (FIG. 7). In some embodiments, the layout design **200**, **400** or **716** is a graphic database system (GDSII) file format.

In operation **604** of method **600A**, the memory array circuit (e.g., memory macro **100A**, memory cell **100B** or **100C**) is manufactured based on layout design **200**, **400** or **716**. In some embodiments, operation **604** of method **600A** comprises manufacturing at least one mask based on the layout design **200**, **400** or **716**, and manufacturing the memory array circuit based on the at least one mask. In some embodiments, the memory array circuit of operation **602** or **604** comprises memory cell **100B** (FIG. 1B), memory cell **100C** (FIG. 1C), IC structure **300** (FIGS. 3A-3I) or IC structure **500** (FIG. 5).

FIG. 6B is a flowchart of a method **600B** of generating a layout design of a memory array circuit in accordance with some embodiments. It is understood that additional operations may be performed before, during, and/or after the method **600B** depicted in FIG. 6B, and that some other processes may only be briefly described herein. In some embodiments, the method **600B** is usable to generate one or more of layout designs **200** or **400** (FIGS. 2A-2C & 4A-4B) of memory cell **100B** (FIG. 1B), memory cell **100C** (FIG. 1C), IC structure **300** (FIGS. 3A-3I) or IC structure **500** (FIG. 5).

In operation **610** of method **600B**, a set of active region layout patterns **212a** or **212b** is generated. In some embodiments, generating the set of active region layout patterns **212a** or **212b** are usable or correspond to fabricating a set of active regions **303** or **305** of integrated circuit **300** or **500**.

In some embodiments, the set of active regions **303** or **305** of integrated circuit **300** or **500** include the active regions of a first pull down transistor (NMOS transistor PD0), a second pull down transistor (NMOS transistor PD1) and a first pass gate transistor (PMOS transistor PG0), a second pass gate transistor (PMOS transistor PG1). In some embodiments, the set of active regions **303** or **305** of integrated circuit **300** or **500** include the active regions of a first pull up transistor (PMOS transistor PU0), a second pull up transistor (PMOS transistor PU1) and a first pass gate transistor (NMOS transistor PG2), a second pass gate transistor (NMOS transistor PG3).

In some embodiments, each of the layout patterns of the set of active region layout patterns **212a**, **212b** is separated from an adjacent layout pattern of the set of active region layout patterns **212a**, **212b** in the second direction Y by a first pitch. In some embodiments, the set of active region layout patterns **212a**, **212b** extend in the first direction X different from the first direction and being located on a first layout level (e.g., active region or well).

In some embodiments, generating the set of active region layout patterns **312a**, **412a** of operation **610** includes generating active region layout pattern **202a1**, active region

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layout pattern **202a2**, active region layout pattern **202b1** and active region layout pattern **202b2**.

In some embodiments, operation **610** includes placing the set of active region layout patterns **312a**, **412a** of on the first layout level (e.g., active region or well). In some embodiments, the first layout level corresponds to the active region of layout design **200** or **400**.

In operation **612**, a set of gate layout patterns **204** is generated. In some embodiments, the set of gate layout patterns **204** corresponds to fabricating a set of gate structures **304** of memory cell **100B-100C** or integrated circuit **300** or **500**.

In some embodiments, the set of gate layout patterns **204** of method **1000** includes one or more of gate layout patterns **204a**, **204b** or **204c**. In some embodiments, operation **612** includes placing the set of gate layout patterns **202** on the second layout level (e.g., POLY).

In operation **614**, a set of metal contact layout patterns **210** is generated. In some embodiments, the set of metal contact layout patterns **210** corresponds to fabricating a set of metal contacts **310** of memory cell **100B-100C** or integrated circuit **300** or **500**.

In some embodiments, in operation **614**, metal contact layout pattern **210a** and **210b** overlaps the cell boundary of layout design **200** or **400** and active region layout pattern **202a**.

In some embodiments, in operation **614**, metal contact layout pattern **210a**, **210b**, **210c** and **210d** are used to manufacture corresponding metal contacts **310a**, **310b**, **310c** and **310d**. In some embodiments, metal contact **310a** is electrically coupled to a source of the first pull down transistor (NMOS transistor PD0). In some embodiments, metal contact **310b** is electrically coupled to a source of the second pull down transistor (NMOS transistor PD1).

In some embodiments, metal contact **310a** is electrically coupled to a source of the first pull up transistor (PMOS transistor PU0). In some embodiments, metal contact **310b** is electrically coupled to a source of the second pull up transistor (PMOS transistor PU1).

In some embodiments, metal contact **310c** is electrically coupled to a source or drain of the first pass gate transistor (PMOS transistor PG0). In some embodiments, metal contact **310d** is electrically coupled to a source or drain of the second pass gate transistor (PMOS transistor PG1).

In some embodiments, metal contact **310c** is electrically coupled to a source or drain of the first pass gate transistor (NMOS transistor PG2). In some embodiments, metal contact **310d** is electrically coupled to a source or drain of the second pass gate transistor (NMOS transistor PG3).

In some embodiments, the set of metal contact layout patterns **210** of method **600B** includes one or more of conductive feature layout patterns **210a**, **210b**, **210c**, **210d**, **210e** or **210f**. In some embodiments, operation **614** includes placing the set of metal contact layout patterns **210** on the third layout level (e.g., MD).

In operation **616**, a first set of conductive feature layout patterns **212** is generated. In some embodiments, the first set of conductive feature layout patterns of operation **616** includes the set of conductive feature layout patterns **212**. In some embodiments, the set of conductive feature layout patterns **212** corresponds to fabricating a set of conductive structures **312** of memory cell **100B-100C** or integrated circuit **300** or **500**.

In some embodiments, the set of conductive feature layout patterns **212** of method **600B** includes one or more of conductive feature layout patterns **212a**, **212b**, **212c**, **212d**, **212e** or **212f**. In some embodiments, operation **616** includes

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placing the set of conductive feature layout patterns **212** on the fourth layout level (e.g., M0).

In operation **618**, a first set of via layout patterns is generated. In some embodiments, the first set of via layout patterns of operation **618** includes set of via layout patterns **214**. In some embodiments, the set of via layout patterns **214** corresponds to fabricating a set of vias **314** of memory cell **100B-100C** or integrated circuit **300** or **500**.

In some embodiments, the set of via layout patterns **214** of method **600B** includes one or more of via layout patterns **214a**, **214b**, **214c**, **214d**, **214e**, **214f**, **214g**, **214h** or **214i**. In some embodiments, operation **618** includes placing the set of vias between the set of metal contact layout patterns **210** and the set of conductive feature layout patterns **212**. In some embodiments, operation **618** includes placing the set of vias between the third layout level (e.g., MD) and the fourth layout level (e.g., M0). In some embodiments, operation **618** includes placing the set of vias **214** on the VG or VD layout level.

In operation **620**, a second set of conductive feature layout patterns is generated. In some embodiments, the second set of conductive feature layout patterns of operation **620** includes the set of conductive feature layout patterns **218**. In some embodiments, the set of conductive feature layout patterns **218** corresponds to fabricating a set of conductive structures **318** of memory cell **100B-100C** or integrated circuit **300** or **500**.

In some embodiments, the set of conductive feature layout patterns **218** of method **600B** includes one or more of conductive feature layout patterns **218a**, **218b**, **218c**, **218d** or **218e**. In some embodiments, operation **620** includes placing the set of conductive feature layout patterns **218** on the fifth layout level (e.g., M1).

In operation **622**, a second set of via layout patterns is generated. In some embodiments, the second set of via layout patterns of operation **622** includes set of via layout patterns **220**. In some embodiments, the set of via layout patterns **220** corresponds to fabricating a set of vias **320** of memory cell **100B-100C** or integrated circuit **300** or **500**.

In some embodiments, the set of via layout patterns **220** of method **600B** includes one or more of via layout patterns **220a**, **220b**, **220c**, **220d** or **220e**. In some embodiments, operation **622** includes placing the set of via layout patterns **220** between the set of conductive feature layout patterns **218** and the set of conductive feature layout patterns **212**.

In some embodiments, operation **622** includes placing the set of vias between the fourth layout level (e.g., M0) and the fifth layout level (e.g., M1). In some embodiments, operation **622** includes placing the set of vias **220** on the V0 layout level.

In operation **624**, a third set of conductive feature layout patterns is generated. In some embodiments, the third set of conductive feature layout patterns of operation **624** includes the set of conductive feature layout patterns **222**. In some embodiments, the set of conductive feature layout patterns **222** corresponds to fabricating a set of conductive structures **322** of memory cell **100B-100C** or integrated circuit **300** or **500**.

In some embodiments, the set of conductive feature layout patterns **222** of method **600B** includes one or more of conductive feature layout patterns **222a**, **222b**, **222c**, **222d**, **222e**, **422b** or **422d**. In some embodiments, operation **624** includes placing the set of conductive feature layout patterns **222** on the sixth layout level (e.g., M2).

In operation **626**, a third set of via layout patterns is generated. In some embodiments, the third set of via layout patterns of operation **626** includes set of via layout patterns

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226. In some embodiments, the set of via layout patterns **226** corresponds to fabricating a set of vias **326** of memory cell **100B-100C** or integrated circuit **300** or **500**.

In some embodiments, the set of via layout patterns **226** of method **600B** includes one or more of via layout patterns **226a**, **226b**, **226c**, **226d** or **226e**. In some embodiments, operation **626** includes placing the set of via layout patterns **226** between the set of conductive feature layout patterns **218** and the set of conductive feature layout patterns **222**. In some embodiments, operation **626** includes placing the set of vias between the fifth layout level (e.g., M1) and the sixth layout level (e.g., M2). In some embodiments, operation **626** includes placing the set of vias **226** on the V1 layout level.

In some embodiments, one or more of layout design **200** or **400** is a standard cell. In some embodiments, one or more of operations **610-626** is not performed.

One or more of the operations of methods **600A-600B** is performed by a processing device configured to execute instructions for manufacturing a memory array circuit, such as memory macro **100A**, a memory cell, such as memory cell **100B** or **100C**, or an IC, such as IC structure **300** or **500**. In some embodiments, one or more operations of methods **600A-600B** is performed using a same processing device as that used in a different one or more operations of methods **600A-600B**. In some embodiments, a different processing device is used to perform one or more operations of methods **600A-600B** from that used to perform a different one or more operations of methods **600A-600B**.

FIG. 7 is a schematic view of a system **700** for designing an IC layout design in accordance with some embodiments. In some embodiments, system **700** generates or places one or more IC layout designs described herein. System **700** includes a hardware processor **702** and a non-transitory, memory **704** encoded with, i.e., storing, the computer program code **706**, i.e., a set of executable instructions. Memory **704** is configured for interfacing with manufacturing machines for producing the integrated circuit. The processor **702** is electrically coupled to the memory **704** via a bus **708**. The processor **702** is also electrically coupled to an I/O interface **710** by bus **708**. A network interface **712** is also electrically connected to the processor **702** via bus **708**. Network interface **712** is connected to a network **714**, so that processor **702** and memory **704** are capable of connecting to external elements via network **714**. The processor **702** is configured to execute the computer program code **706** encoded in the memory **704** in order to cause system **700** to be usable for performing a portion or all of the operations as described in method **600A** or **600B**.

In some embodiments, the processor **702** is a central processing unit (CPU), a multi-processor, a distributed processing system, an application specific integrated circuit (ASIC), and/or a suitable processing unit.

In some embodiments, the memory **704** is an electronic, magnetic, optical, electromagnetic, infrared, and/or a semiconductor system (or apparatus or device). For example, the memory **704** includes a semiconductor or solid-state memory, a magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), a rigid magnetic disk, and/or an optical disk. In some embodiments, memory **704** is a computer readable storage medium. In some embodiments using optical disks, the memory **704** includes a compact disk-read only memory (CD-ROM), a compact disk-read/write (CD-R/W), and/or a digital video disc (DVD).

In some embodiments, the storage medium **704** stores the computer program code **706** configured to cause system **700** to perform method **600A** or **600B**. In some embodiments,

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the storage medium **704** also stores information needed for performing method **600A** or **600B** as well as information generated during performing method **600A** or **600B**, such as layout design **716** and user interface **718**, and/or a set of executable instructions to perform the operation of method **600A** or **600B**. In some embodiments, layout design **716** comprises one or more of layout designs **200** or **400**.

In some embodiments, the storage medium **704** stores instructions (e.g., computer program code **706**) for interfacing with manufacturing machines. The instructions (e.g., computer program code **706**) enable processor **702** to generate manufacturing instructions readable by the manufacturing machines to effectively implement method **600A** or **600B** during a manufacturing process.

System **700** includes I/O interface **710**. I/O interface **710** is coupled to external circuitry. In some embodiments, I/O interface **710** includes a keyboard, keypad, mouse, trackball, trackpad, and/or cursor direction keys for communicating information and commands to processor **702**.

System **700** also includes network interface **712** coupled to the processor **702**. Network interface **712** allows system **700** to communicate with network **714**, to which one or more other computer systems are connected. Network interface **712** includes wireless network interfaces such as BLUETOOTH, WIFI, WIMAX, GPRS, or WCDMA; or wired network interface such as ETHERNET, USB, or IEEE-1394. In some embodiments, method **600A** or **600B** is implemented in two or more systems **700**, and information such as layout design, and user interface are exchanged between different systems **700** by network **714**.

System **700** is configured to receive information related to a layout design through I/O interface **710** or network interface **712**. The information is transferred to processor **702** by bus **708** to determine a layout design for producing an IC structure, such as IC structure **300** or **500**. The layout design is then stored in memory **704** as layout design **716**. System **700** is configured to receive information related to a user interface through I/O interface **710** or network interface **712**. The information is stored in memory **704** as user interface **718**.

In some embodiments, method **600A** or **600B** is implemented as a standalone software application for execution by a processor. In some embodiments, method **600A** or **600B** is implemented as a software application that is a part of an additional software application. In some embodiments, method **600A** or **600B** is implemented as a plug-in to a software application. In some embodiments, method **600A** or **600B** is implemented as a software application that is a portion of an EDA tool. In some embodiments, method **600A** or **600B** is implemented as a software application that is used by an EDA tool. In some embodiments, the EDA tool is used to generate a layout of the integrated circuit device. In some embodiments, the layout is stored on a non-transitory computer readable medium or memory **704**. In some embodiments, the layout is generated using a tool such as VIRTUOSO® available from CADENCE DESIGN SYSTEMS, Inc., or another suitable layout generating tool. In some embodiments, the layout is generated based on a netlist which is created based on the schematic design. In some embodiments, method **600A** is implemented by a manufacturing device to manufacture an integrated circuit (e.g., memory cell **100B** or **100C**, IC structure **300** or **500**) using a set of masks manufactured based on one or more layout designs (e.g., layout design **200** or **400**) generated by system **700**.

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System **700** of FIG. 7 generates layout designs (e.g., layout design **200** or **400**) of memory macro **100A**, memory cell **100B** or **100C** or IC structure **300** or **500** that are smaller than other approaches.

FIG. 8 is a block diagram of an integrated circuit (IC) manufacturing system **800**, and an IC manufacturing flow associated therewith, in accordance with at least one embodiment of the present disclosure.

In FIG. 8, IC manufacturing system **800** includes entities, such as a design house **820**, a mask house **830**, and an IC manufacturer/fabricator ("fab") **840**, that interact with one another in the design, development, and manufacturing cycles and/or services related to manufacturing an IC device **860**. The entities in system **800** are connected by a communications network. In some embodiments, the communications network is a single network. In some embodiments, the communications network is a variety of different networks, such as an intranet and the Internet. The communications network includes wired and/or wireless communication channels. Each entity interacts with one or more of the other entities and provides services to and/or receives services from one or more of the other entities. In some embodiments, two or more of design house **820**, mask house **830**, and IC fab **840** is owned by a single larger company. In some embodiments, two or more of design house **820**, mask house **830**, and IC fab **840** coexist in a common facility and use common resources.

Design house (or design team) **820** generates an IC design layout **822**. IC design layout **822** includes various geometrical patterns designed for an IC device **860**. The geometrical patterns correspond to patterns of metal, oxide, or semiconductor layers that make up the various components of IC device **860**. The various layers combine to form various IC features. For example, a portion of IC design layout **822** includes various IC features, such as an active region, gate electrode, source electrode and drain electrode, metal lines or vias of an interlayer interconnection, and openings for bonding pads, to be formed in a semiconductor substrate (such as a silicon wafer) and various material layers disposed on the semiconductor substrate. Design house **820** implements a proper design procedure to form IC design layout **822**. The design procedure includes one or more of logic design, physical design or place and route. IC design layout **822** is presented in one or more data files having information of the geometrical patterns. For example, IC design layout **822** can be expressed in a GDSII file format or DFII file format.

Mask house **830** includes data preparation **832** and mask fabrication **834**. Mask house **830** uses IC design layout **822** to manufacture one or more masks to be used for fabricating the various layers of IC device **860** according to IC design layout **822**. Mask house **830** performs mask data preparation **832**, where IC design layout **822** is translated into a representative data file ("RDF"). Mask data preparation **832** provides the RDF to mask fabrication **834**. Mask fabrication **834** includes a mask writer. A mask writer converts the RDF to an image on a substrate, such as a mask (reticle) or a semiconductor wafer. The IC design layout **822** is manipulated by mask data preparation **832** to comply with particular characteristics of the mask writer and/or requirements of IC fab **840**. In FIG. 8, mask data preparation **832** and mask fabrication **834** are illustrated as separate elements. In some embodiments, mask data preparation **832** and mask fabrication **834** can be collectively referred to as mask data preparation.

In some embodiments, mask data preparation **832** includes optical proximity correction (OPC) which uses

lithography enhancement techniques to compensate for image errors, such as those that can arise from diffraction, interference, other process effects and the like. OPC adjusts IC design layout **822**. In some embodiments, mask data preparation **832** includes further resolution enhancement techniques (RET), such as off-axis illumination, sub-resolution assist features, phase-shifting masks, other suitable techniques, and the like or combinations thereof. In some embodiments, inverse lithography technology (ILT) is also used, which treats OPC as an inverse imaging problem.

In some embodiments, mask data preparation **832** includes a mask rule checker (MRC) that checks the IC design layout that has undergone processes in OPC with a set of mask creation rules which contain certain geometric and/or connectivity restrictions to ensure sufficient margins, to account for variability in semiconductor manufacturing processes, and the like. In some embodiments, the MRC modifies the IC design layout **822** to compensate for limitations during mask fabrication **834**, which may undo part of the modifications performed by OPC in order to meet mask creation rules.

In some embodiments, mask data preparation **832** includes lithography process checking (LPC) that simulates processing that will be implemented by IC fab **840** to fabricate IC device **860**. LPC simulates this processing based on IC design layout **822** to create a simulated manufactured device, such as IC device **860**. The processing parameters in LPC simulation can include parameters associated with various processes of the IC manufacturing cycle, parameters associated with tools used for manufacturing the IC, and/or other aspects of the manufacturing process. LPC takes into account various factors, such as aerial image contrast, depth of focus ("DOF"), mask error enhancement factor ("MEEF"), other suitable factors, and the like or combinations thereof. In some embodiments, after a simulated manufactured device has been created by LPC, if the simulated device is not close enough in shape to satisfy design rules, OPC and/or MRC are repeated to further refine IC design layout **822**.

It should be understood that the above description of mask data preparation **832** has been simplified for the purposes of clarity. In some embodiments, data preparation **832** includes additional features such as a logic operation (LOP) to modify the IC design layout **822** according to manufacturing rules. Additionally, the processes applied to IC design layout **822** during data preparation **832** may be executed in a variety of different orders.

After mask data preparation **832** and during mask fabrication **834**, a mask or a group of masks are fabricated based on the modified IC design layout. In some embodiments, an electron-beam (e-beam) or a mechanism of multiple e-beams is used to form a pattern on a mask (photomask or reticle) based on the modified IC design layout. The mask can be formed in various technologies. In some embodiments, the mask is formed using binary technology. In some embodiments, a mask pattern includes opaque regions and transparent regions. A radiation beam, such as an ultraviolet (UV) beam, used to expose the image sensitive material layer (e.g., photoresist) which has been coated on a wafer, is blocked by the opaque region and transmits through the transparent regions. In one example, a binary mask includes a transparent substrate (e.g., fused quartz) and an opaque material (e.g., chromium) coated in the opaque regions of the mask. In another example, the mask is formed using a phase shift technology. In the phase shift mask (PSM), various features in the pattern formed on the mask are configured to have proper phase difference to enhance the

resolution and imaging quality. In various examples, the phase shift mask can be attenuated PSM or alternating PSM. The mask(s) generated by mask fabrication **834** is used in a variety of processes. For example, such a mask(s) is used in an ion implantation process to form various doped regions in the semiconductor wafer, in an etching process to form various etching regions in the semiconductor wafer, and/or in other suitable processes.

IC fab **840** is an IC fabrication business that includes one or more manufacturing facilities for the fabrication of a variety of different IC products. In some embodiments, IC Fab **840** is a semiconductor foundry. For example, there may be a manufacturing facility for the front end fabrication of a plurality of IC products (front-end-of-line (FEOL) fabrication), while a second manufacturing facility may provide the back end fabrication for the interconnection and packaging of the IC products (back-end-of-line (BEOL) fabrication), and a third manufacturing facility may provide other services for the foundry business.

IC fab **840** uses the mask (or masks) fabricated by mask house **830** to fabricate IC device **860**. Thus, IC fab **840** at least indirectly uses IC design layout **822** to fabricate IC device **860**. In some embodiments, a semiconductor wafer **842** is fabricated by IC fab **840** using the mask (or masks) to form IC device **860**. Semiconductor wafer **842** includes a silicon substrate or other proper substrate having material layers formed thereon. Semiconductor wafer further includes one or more of various doped regions, dielectric features, multilevel interconnects, and the like (formed at subsequent manufacturing steps).

One aspect of this description relates to a memory cell. In some embodiments, the memory cell includes a first pull up transistor, and a first pass gate transistor coupled to the first pull up transistor. In some embodiments, the memory cell further includes a second pull up transistor having a first active region extending in a first direction, and the first active region being located on a first level. In some embodiments, the memory cell further includes a second pass gate transistor having a second active region extending in the first direction, the second active region being located on the first level, and being separated from the first active region in a second direction different from the first direction, and the second active region being adjacent to the first active region, the second pass gate transistor being coupled to the second pull up transistor. In some embodiments, the memory cell further includes a first metal contact extending in the second direction, and extending from the first active region to the second active region, the first metal contact being located on a second level different from the first level, the first metal contact electrically coupling a drain of the second pull up transistor to a drain of the second pass gate transistor. In some embodiments, the first pass gate transistor, the second pass gate transistor, the first pull up transistor and the second pull up transistor are part of a four transistor (4T) memory cell.

Still another aspect of this disclosure relates to a memory cell. In some embodiments, the memory cell includes a first pull down transistor having a first active region extending in a first direction, and the first active region being located on a first level. In some embodiments, the memory cell further includes a first pass gate transistor having a second active region extending in the first direction, the second active region being located on the first level, and being separated from the first active region in a second direction different from the first direction. In some embodiments, the memory cell further includes a second pull down transistor having a third active region extending in the first direction, the third

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active region being located on the first level, and being separated from the first active region in the first direction. In some embodiments, the memory cell further includes a second pass gate transistor having a fourth active region extending in the first direction, the fourth active region being located on the first level, being separated from the third active region in the second direction, and being separated from the second active region in the first direction. In some embodiments, the memory cell further includes a first metal contact extending in the second direction, and extending from the first active region to the second active region, the first metal contact being located on a second level different from the first level, the first metal contact electrically coupling a drain of the first pull down transistor to a drain of the first pass gate transistor. In some embodiments, the first pass gate transistor, the second pass gate transistor, the first pull down transistor and the second pull down transistor are part of a four transistor (4T) memory cell.

Yet another aspect of this description relates to a memory cell. In some embodiments, the memory cell includes a first pull down transistor, and a first pass gate transistor coupled to the first pull down transistor. In some embodiments, the memory cell further includes a second pull down transistor having a first active region extending in a first direction, and the first active region being located on a first level. In some embodiments, the memory cell further includes a second pass gate transistor having a second active region extending in the first direction, the second active region being located on the first level, and being separated from the first active region in a second direction different from the first direction, the second pass gate transistor coupled to the second pull down transistor. In some embodiments, the memory cell further includes a first metal contact extending in the second direction, and extending from the first active region to the second active region, the first metal contact being located on a second level different from the first level, the first metal contact electrically coupling a drain of the second pull down transistor to a drain of the second pass gate transistor. In some embodiments, the first pass gate transistor, the second pass gate transistor, the first pull down transistor and the second pull down transistor are part of a four transistor (4T) memory cell.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A memory cell comprising:

- a first pull up transistor;
- a first pass gate transistor coupled to the first pull up transistor;
- a second pull up transistor having a first active region extending in a first direction, and the first active region being located on a first level;
- a second pass gate transistor having a second active region extending in the first direction, the second active region being located on the first level, and being separated from the first active region in a second

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direction different from the first direction, and the second active region being adjacent to the first active region, the second pass gate transistor being coupled to the second pull up transistor; and

- a first metal contact extending in the second direction, and extending from the first active region to the second active region, the first metal contact being located on a second level different from the first level, the first metal contact electrically coupling a drain of the second pull up transistor to a drain of the second pass gate transistor;

wherein the first pass gate transistor, the second pass gate transistor, the first pull up transistor and the second pull up transistor are part of a four transistor (4T) memory cell.

2. The memory cell of claim 1, further comprising:

- a second metal contact extending in the second direction, overlapping the first active region, and being located on the second level, the second metal contact being electrically coupled to a source of the second pull down transistor and a voltage supply; and
- a third metal contact extending in the second direction, overlapping the second active region, and being located on the second level, the third metal contact being electrically coupled to a source of the second pass gate transistor.

3. The memory cell of claim 2, further comprising:

- a set of gates comprising:
 - a first gate extending in the second direction, overlapping the first active region, and being located on the second level; and
 - a second gate extending in the second direction, overlapping the second active region being located on the second level, and being separated from the first gate in the second direction.

4. The memory cell of claim 3, further comprising:

- a first set of conductive structures extending in the first direction and overlapping at least the first active region, the second active region or the set of gates, each conductive structure of the first set of conductive structures being separated from an adjacent conductive structure of the first set of conductive structures in at least the first direction or the second direction, and being located on a third level different from the first level and the second level.

5. The memory cell of claim 4, further comprising:

- a first set of vias between the first set of conductive structures and the first active region and the second active region, the first set of vias coupling the first set of conductive structures to at least the first active region, the second active region or the set of gates, and at least one via of the first set of vias being located where at least one conductive structure of the first set of conductive structures overlaps at least one of the first active region, the second active region or the set of gates.

6. The memory cell of claim 5, further comprising:

- a second set of conductive structures extending in the second direction and overlapping at least the first active region, the second active region or the first set of conductive structures, each conductive structure of the second set of conductive structures being separated from an adjacent structure of the second set of conductive structures in the first direction, and being located on a fourth level different from the first level, the second level and the third level; and

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- a second set of vias between the second set of conductive structures and the first set of conductive structures, the second set of vias coupling the second set of conductive structures to the first set of conductive structures, and at least one via of the second set of vias being located where at least one conductive structure of the second set of conductive structures overlaps at least one of the first set of conductive structures. 5
7. The memory cell of claim 6, further comprising:
- a third set of conductive structures extending in the first direction and overlapping at least the second set of conductive structures, each conductive structure of the third set of conductive structures being separated from an adjacent structure of the third set of conductive structures in the second direction, and being located on a fifth level different from the first level, the second level, the third level and the fourth level; and 10
- a third set of vias between the third set of conductive structures and the second set of conductive structures, the third set of vias coupling the third set of conductive structures to the second set of conductive structures, and at least one via of the third set of vias being located where at least one conductive structure of the third set of conductive structures overlaps at least one conductive structure of the second set of conductive structures. 15 25
8. A memory cell comprising:
- a first pull down transistor having a first active region extending in a first direction, and the first active region being located on a first level;
- a first pass gate transistor having a second active region extending in the first direction, the second active region being located on the first level, and being separated from the first active region in a second direction different from the first direction; 30
- a second pull down transistor having a third active region extending in the first direction, the third active region being located on the first level, and being separated from the first active region in the first direction; 35
- a second pass gate transistor having a fourth active region extending in the first direction, the fourth active region being located on the first level, being separated from the third active region in the second direction, and being separated from the second active region in the first direction; and 40
- a first metal contact extending in the second direction, and extending from the first active region to the second active region, the first metal contact being located on a second level different from the first level, the first metal contact electrically coupling a drain of the first pull down transistor to a drain of the first pass gate transistor, 45 50
- wherein the first pass gate transistor, the second pass gate transistor, the first pull down transistor and the second pull down transistor are part of a four transistor (4T) memory cell. 55
9. The memory cell of claim 8, further comprising:
- a set of gates comprising:
- a first gate extending in the second direction, overlapping the first active region or the second active region, and being located on the second level; and 60
- a second gate extending in the second direction, overlapping the third active region or the fourth active region, being located on the second level, and being separated from the first gate in the second direction.
10. The memory cell of claim 9, further comprising: 65
- a first set of conductive structures extending in the first direction and overlapping at least the first active region,

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- the second active region, the third active region, the fourth active region or the set of gates, each conductive structure of the first set of conductive structures being separated from an adjacent conductive structure of the first set of conductive structures in at least the first direction or the second direction, and being located on a third level different from the first level and the second level; and
- a first set of vias between the first set of conductive structures and the first active region and the second active region, the first set of vias coupling the first set of conductive structures to at least the first active region, the second active region, the third active region, the fourth active region or the set of gates, and at least one via of the first set of vias being located where at least one conductive structure of the first set of conductive structures overlaps at least one of the first active region or the second active region.
11. The memory cell of claim 10, further comprising:
- a second set of conductive structures extending in the second direction and overlapping at least the first active region, the second active region, the third active region, the fourth active region or the first set of conductive structures, each conductive structure of the second set of conductive structures being separated from an adjacent structure of the second set of conductive structures in the first direction, and being located on a fourth level different from the first level, the second level and the third level; and
- a second set of vias between the second set of conductive structures and the first set of conductive structures, the second set of vias coupling the second set of conductive structures to the first set of conductive structures, and at least one via of the second set of vias being located where at least one conductive structure of the second set of conductive structures overlaps at least one of the first set of conductive structures.
12. The memory cell of claim 11, further comprising:
- a third set of conductive structures extending in the first direction and overlapping at least the second set of conductive structures, each conductive structure of the third set of conductive structures being separated from an adjacent structure of the third set of conductive structures in the second direction, and being located on a fifth level different from the first level, the second level, the third level and the fourth level; and
- a third set of vias between the third set of conductive structures and the second set of conductive structures, the third set of vias coupling the third set of conductive structures to the second set of conductive structures, and at least one via of the third set of vias being located where at least one conductive structure of the third set of conductive structures overlaps at least one conductive structure of the second set of conductive structures.
13. The memory cell of claim 12, wherein
- the first set of vias comprises:
- a first via of the first set of vias electrically coupling the first gate to a first conductive structure of the first set of conductive structures; and
- a second via of the first set of vias electrically coupling the second gate to the first conductive structure of the first set of conductive structures;
- the second set of vias comprises:
- a first via of the second set of vias electrically coupling the first conductive structure of the first set of conductive structures to a first conductive structure of the second set of conductive structures; and

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- a second via of the second set of vias electrically coupling the first conductive structure of the first set of conductive structures to a second conductive structure of the second set of conductive structures; and
 the third set of vias comprises:
 a first via of the third set of vias electrically coupling a first conductive structure of the third set of conductive structures to a third conductive structure of the second set of conductive structures, the first conductive structure of the third set of conductive structures corresponding to a power supply rail coupled to a voltage supply or a reference supply.
- 14.** A memory cell comprising:
 a first pull down transistor;
 a first pass gate transistor coupled to the first pull down transistor;
 a second pull down transistor having a first active region extending in a first direction, and the first active region being located on a first level;
 a second pass gate transistor having a second active region extending in the first direction, the second active region being located on the first level, and being separated from the first active region in a second direction different from the first direction, the second pass gate transistor coupled to the second pull down transistor;
 a first metal contact extending in the second direction, and extending from the first active region to the second active region, the first metal contact being located on a second level different from the first level, the first metal contact electrically coupling a drain of the second pull down transistor to a drain of the second pass gate transistor; and
 wherein the first pass gate transistor, the second pass gate transistor, the first pull down transistor and the second pull down transistor are part of a four transistor (4T) memory cell.
- 15.** The memory cell of claim **14**, further comprising:
 a second metal contact extending in the second direction, overlapping the first active region, and being located on the second level, the second metal contact being electrically coupled to a source of the second pull down transistor and a reference voltage supply; and
 a third metal contact extending in the second direction, overlapping the second active region, and being located on the second level, the third metal contact being electrically coupled to a source of the second pass gate transistor.
- 16.** The memory cell of claim **15**, further comprising:
 a set of gates comprising:
 a first gate extending in the second direction, overlapping the first active region, and being located on the second level; and
 a second gate extending in the second direction, overlapping the second active region being located on the second level, and being separated from the first gate in the second direction.
- 17.** The memory cell of claim **16**, further comprising:
 a first set of conductive structures extending in the first direction and overlapping at least the first active region, the second active region or the set of gates, each conductive structure of the first set of conductive structures being separated from an adjacent conductive structure of the first set of conductive structures in at

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- least the first direction or the second direction, and being located on a third level different from the first level and the second level; and
 a first set of vias between the first set of conductive structures and at least the first active region, the second active region or the set of gates, the first set of vias coupling the first set of conductive structures to at least the first active region, the second active region or the set of gates, and at least one via of the first set of vias being located where at least one conductive structure of the first set of conductive structures overlaps at least one of the first active region, the second active region or the set of gates.
- 18.** The memory cell of claim **17**, further comprising:
 a second set of conductive structures extending in the second direction and overlapping at least the first active region, the second active region or the first set of conductive structures, each conductive structure of the second set of conductive structures being separated from an adjacent structure of the second set of conductive structures in the first direction, and being located on a fourth level different from the first level, the second level and the third level; and
 a second set of vias between the second set of conductive structures and the first set of conductive structures, the second set of vias coupling the second set of conductive structures to the first set of conductive structures, and at least one via of the second set of vias being located where at least one conductive structure of the second set of conductive structures overlaps at least one of the first set of conductive structures.
- 19.** The memory cell of claim **18**, further comprising:
 a third set of conductive structures extending in the first direction and overlapping at least the second set of conductive structures, each conductive structure of the third set of conductive structures being separated from an adjacent structure of the third set of conductive structures in the second direction, and being located on a fifth level different from the first level, the second level, the third level and the fourth level; and
 a third set of vias between the third set of conductive structures and the second set of conductive structures, the third set of vias coupling the third set of conductive structures to the second set of conductive structures, and at least one via of the third set of vias being located where at least one conductive structure of the third set of conductive structures overlaps at least one conductive structure of the second set of conductive structures.
- 20.** The memory cell of claim **19**, wherein
 the first set of vias comprises:
 a first via of the first set of vias electrically coupling the first gate to a first conductive structure of the first set of conductive structures; and
 a second via of the first set of vias electrically coupling the second gate to a second conductive structure of the first set of conductive structures; and
 the second set of vias comprises:
 a first via of the second set of vias electrically coupling the second conductive structure of the first set of conductive structures to a first conductive structure of the second set of conductive structures, the first conductive structure of the second set of conductive structures corresponding to a first word line.

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